

دائرة التمكين الحكومي
DEPARTMENT OF GOVERNMENT ENABLEMENT



AI in Government

Government Offsite

Monday, May 20 – Tuesday, May 21, 2024

Day 1 Presentation

Opening Remarks

H.E. Ahmed Tamim Al Kuttab

Chairman of Department of Government Enablement



An AI-native Government

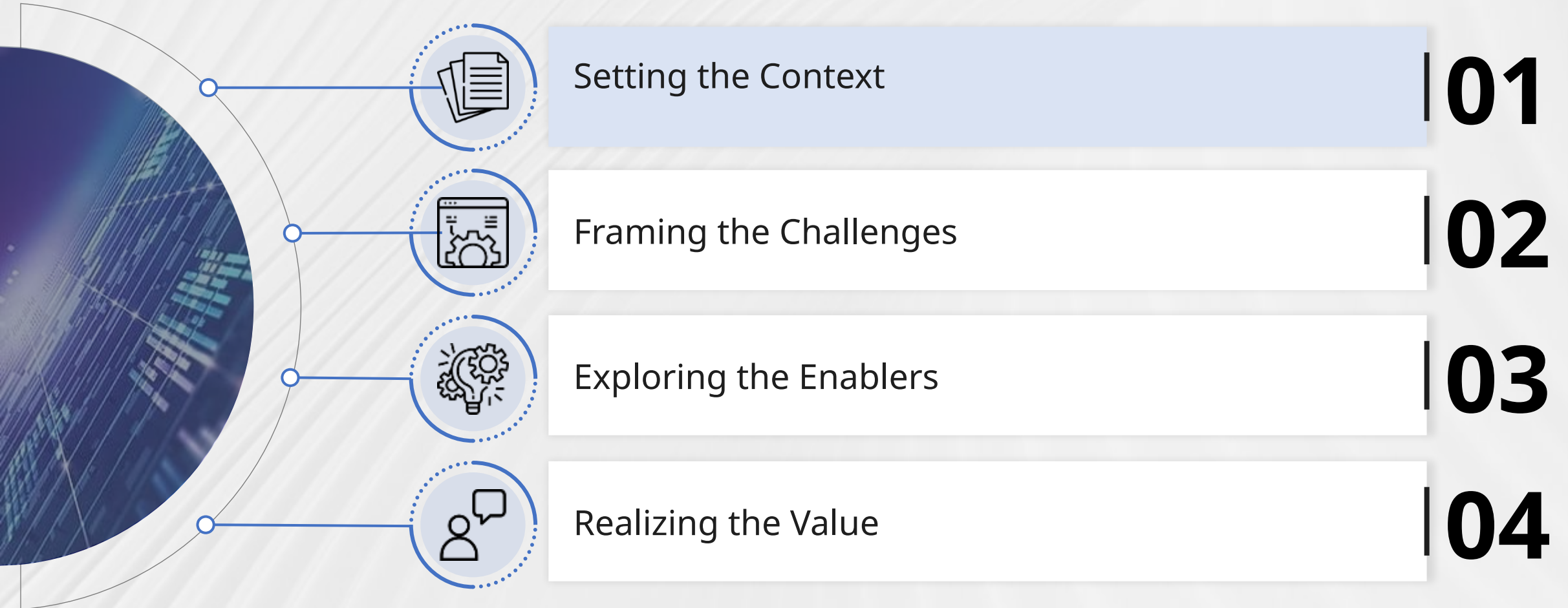


Driven by value creation and guided by ethical-regulatory framework



Accelerated by a collaborative ecosystem of public and private sector players

Day 1 Agenda



Day 1 Agenda

Topic	Location	Timing	Duration	Presentation	Speakers
Introduction	Auditorium	9:00 – 9:15	15min	Welcome Message	EC Secretary General - TBC
	AI Studio	9:15 – 9:20	5min	Agenda & Objectives	H.E. Ahmed Al Kuttab
Setting the Context	AI Studio	9:20 – 9:40	20min	Unlock the full potential of AI in UAE	H.E. Faisal Al Bannai
	AI Studio	9:40 – 10:00	20min	Demystifying AI	Professor Eric Xing
Framing the Challenges	AI Studio	10:00 – 10:20	20min	Challenges	H.E. Saleh Hamed
Exploring the Enablers	AI Studio	10:20 – 10:40	20min	AI Tech, Data & Security	H.E. Peng Xiao
	AI Studio	10:40 – 11:00	20min	Operating model & Ecosystem	Melvina Tarazi
Break	Break	11:00 – 11:15	15min	Break	
	Auditorium	11:15 – 12:30	1h15min	Presentations: Education, Obesity and AI (morning recap)	
	Break	12:30 – 13:30	1h	Prayer & Lunch break	
Exploring the Enablers	AI Studio	13:30 – 14:00	30min	AI Talent and Org Design	H.E. Saleh Hamed
Value Creation	AI Studio	14:00 – 14:55	55min	AI Accelerator Use case identification & prioritization, Use case selection TAMM 3.0 Demo	Melvina Tarazi H.E. Mohamed Al Askar
	AI Studio	14:55-15:00	5min	“AI Big Ideas”	H.E. Saleh Hamed
Break	Break	15:00 – 15:15	15min	Break	
	Auditorium	15:15 – 16:30	1h15min	Presentations: Education, Obesity and AI (afternoon recap)	



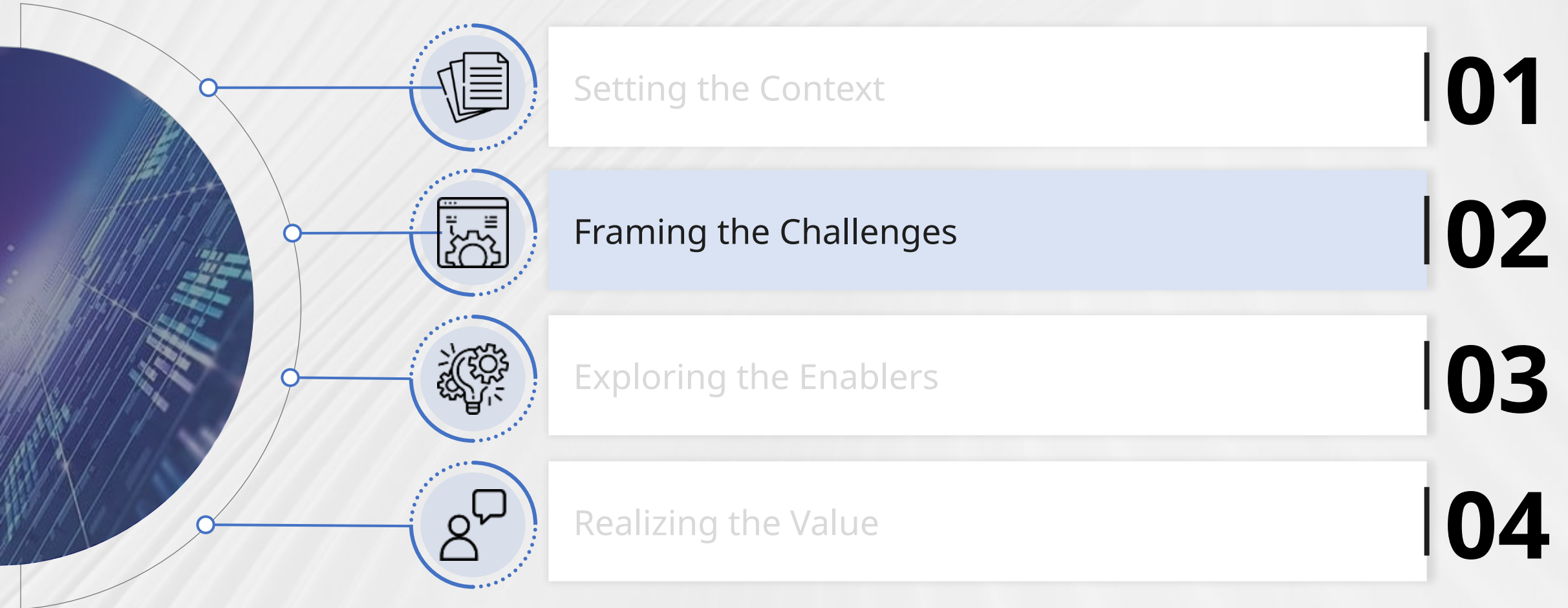
Becoming an AI-native Government



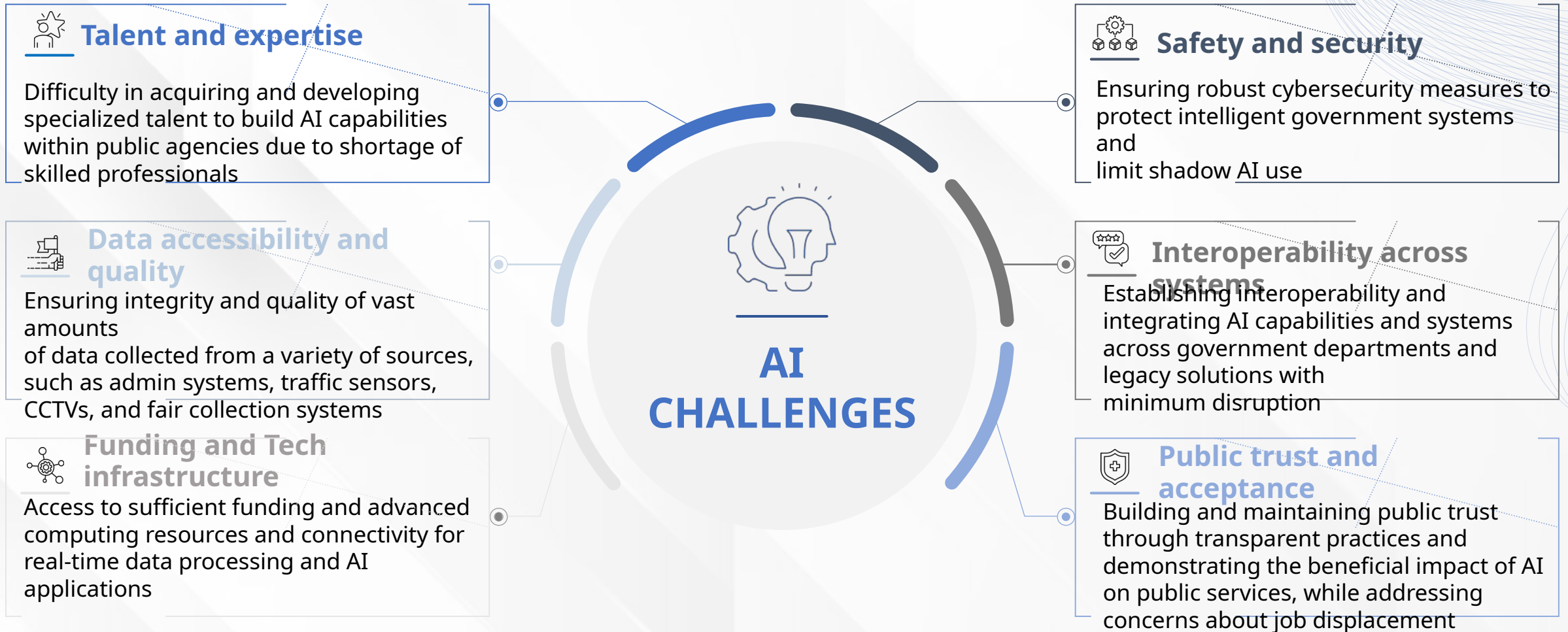
Unlock the full potential of AI in UAE

Demystifying AI

Agenda



Governments typically encounter a series of challenges when implementing AI use cases around 6 key topics



ADGEs highlighted several challenges in advancing their AI agenda



Data Accessibility and Quality

67% of ADGEs reported **having incomplete** or **emerging data strategies**

30% of ADGEs have implemented **data catalog tools**

48% of ADGEs self rate their **data quality as fair** – at best



Cloud Readiness

35% of ADGEs **have 95% to 100% of workloads on- Premise**

33% of ADGEs have over **90% of workloads on cloud**

91% of ADGEs require **cloud services** to be able to activate **92% of the prioritized AI use cases**



Talent / Culture

86% of ADGEs have resource deficiencies and facing **talent acquisition and retention challenges**

63% reported having **no or limited data upskilling initiatives for their staff**

86% of ADGEs have resource deficiencies and facing **talent acquisition and retention challenges**



Operating Model Maturity

90% of ADGEs expect **support** from DGE on **strategy development**

67% of ADGEs **have no or ad-hoc data governance structures**

53% and **43%** reported **unclear accountability** and **inadequate enforcement** as a top challenges



Security & Ecosystem

Inconsistent or non-existent security controls across the **55** ADGEs

Many of the **data sets** not classified with enough detail

100% of ADGEs require **ecosystem support** to be able to activate 100% of the prioritized AI use cases

Challenges we face in delivering on the AI Agenda and implementing AI Use Cases

USE CASES

Are you able to articulate the value of the Use Cases? Identifying critical components to deliver.

BUSINESS ENGAGEMENT

How can this be a business and not an IT Agenda?

AI PRODUCTS

Challenges in identifying appropriate offering, choices of build vs buy? Interoperating with legacy systems

TECH, DATA & SECURITY

What are the challenges you are facing in activating tech stack, data accessibility and quality? How much of your systems are legacy?



PARTNERSHIPS

Are you able to identify the right partner and what challenges are you facing?

FUNDING

Are your challenged by access to funding?

TALENT

Do you have AI talent? Are you able to attract?

STRATEGY AND GOVERNANCE

What do you think are the challenges in terms of Strategy and Governance?

Let's divide into 3 groups to address the challenges

Group 1

STRATEGY AND GOVERNANCE

What support do you required in terms of Strategy and Governance?

FUNDING

Are your challenged by access to funding?

TALENT

Do you have AI talent? Are you able to attract?

Group 2

USE CASES

Are you able to articulate the value of the Use Cases? Identifying critical components to deliver.

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AI PRODUCTS

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TECH, DATA & SECURITY

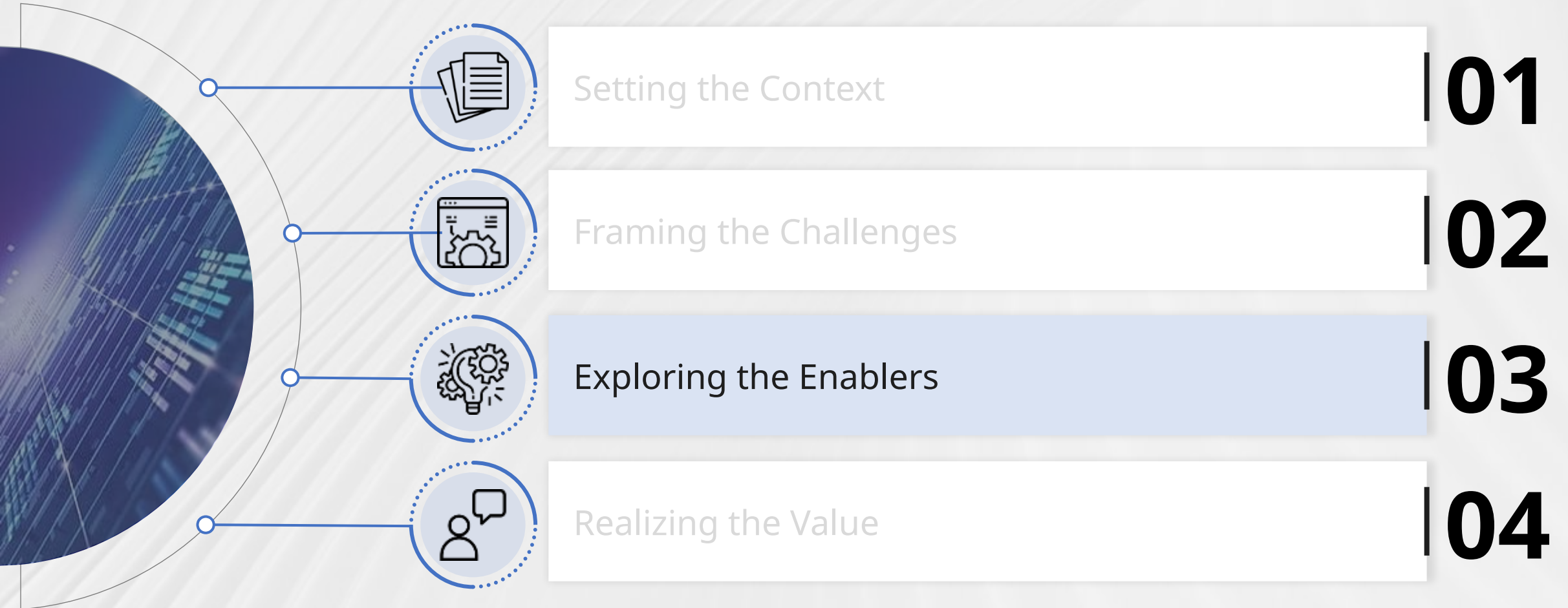
What are the challenges you are facing in activating tech stack, data accessibility and quality? How much of your systems are legacy?

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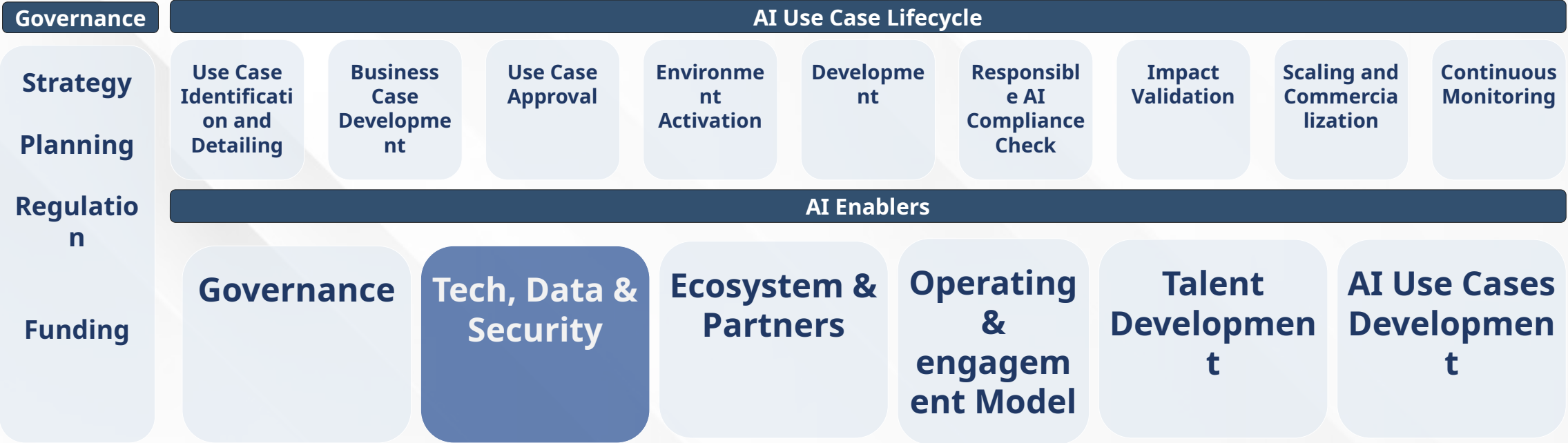
Group 1	Group 2	Group 3
Facilitator Ziad Mohamed	Facilitator Saleh Hamed	Facilitator Melvina Tarazi
Professor Eric Xing	Ahmed Al Kuttub	Awaidha Al Marar
Badr Al Olama (DED)	Faisal Al Bannai	Mohammed Asker (DGE)
Abdulla Al Qemzi (SCAD)	Abdulla Marzooqi (DMT)	Mazen Dhamani (DCD)
Marwan Othman (ADEK)	Saeed Al Fazari (DCT)	Nasser Farah (DMT)
Ibrahim Al Jallaf (DOH)	Nasser AlRahma (ADEO)	Peng Xiao



Agenda



To address the challenges and achieve value from AI investments



Governments are tackling the AI Tech, Data & Security hurdles to meet their AI vision



- **Chinese Government opens data platforms, Public data company platforms (Alibaba, Tencent, Baidu, etc.)**



- **Digital Infrastructure Act** to standardize the requirements for infrastructure (e.g. data centers) to enforce energy efficiency, compliance and security
- **SNDG will invest \$500M in high-performance computer** to support AI capability



- **3 of Big Five tech companies in Silicon Valley**, generating vast amounts of data
- Use of data by Big Tech to **train complex AI models & leverage insights** to drive scientific and commercial advancement



- France 2030 AI Program has **funded the AI-dedicated supercomputer "Jean Zay"** to meet the future requirements of AI computation and innovation
- Government and academic institutions collaborated to set up **CASD Secure Data Hub**



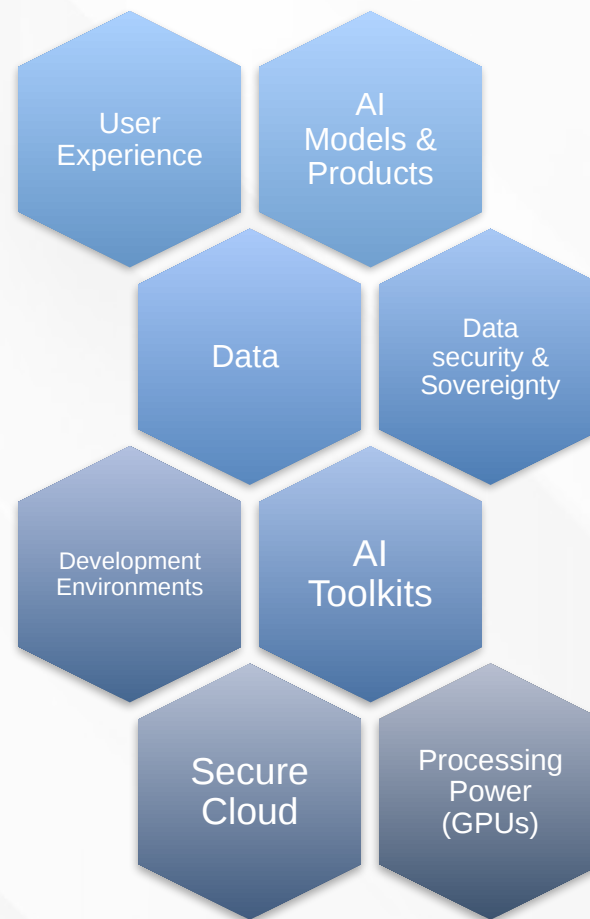
- MCIT¹ launched an **AI training programme⁶** to upskill engineering workforce and the **Artificial Intelligence Challenge** to encourage Saudi youth to drive innovation in AI

How do we address the Tech, Data & Security needs and challenges

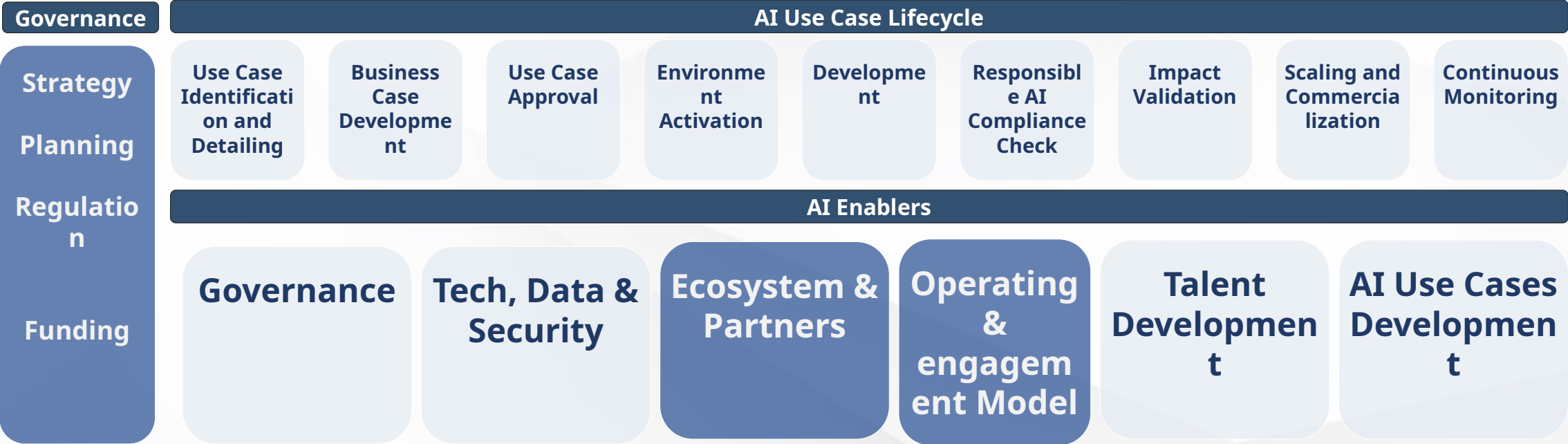
What do we have in the Abu Dhabi ecosystem to address the build-up of the required secure AI Tech Stack?

Is our data secure?

What resource constraints exist?



To address the challenges and achieve value from AI investments



Let's explore some of the other enablers

Governments are tackling the operating model hurdles to meet their AI vision



- **Government programs** such as the National Key Research and Development Program strategically allocate funding to target sectors, including AI



- Govt. established **AI Singapore program** to bring together researchers and companies
- **Industry is bridged with RnD centers** via platforms like NLP Hub, Syneros & CV Hub



- **Universities with dedicated research groups** & laboratories, e.g., Stanford Human-Centered AI Institute (HAI) to **study augmenting human capabilities through AI** – over 13\$M committed to early-stage AI research at Stanford till 2024



- National AI Strategy **dedicated a budget of € 700 million** for National AI Research Program (PNRIA) led by INRIA
- **Multiple RnD Centers** established e.g., SCAI, Hi! PARIS CENTER, PR[AI]RIE, DATAIA

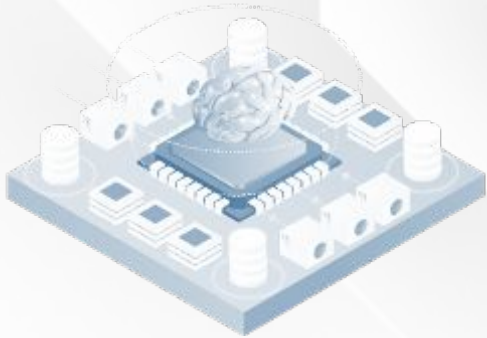


- **KSA National Strategy for Data & Artificial Intelligence** launched in 2020 to drive development and adoption of innovative AI and builds an enabling ecosystem
- **Saudi Data & AI Authority** (SDAIA) oversees implementation of strategy

We have explored 4 Archetypes to advance our thinking

Overview	w	AI INNOVATION OFFICE1	AI ACCELERATOR OFFICE2
		AI COE3	AI HUB FOR PUBLIC-PRIVATE INNOVATION4
Objectives		Internal unit dedicated to integrating AI technologies into its operations and services, serving as a model for digital transformation across the government • Enhance operational efficiency through AI integration. • Improve service delivery to the public and other government entities • Foster a culture of innovation and technological curiosity • Develop and showcase model AI projects that can be replicated	A government-wide office that act as a cornerstone for AI initiatives across the AD, tasked with steering the AI agenda, facilitating the adoption of AI tech, and ensuring governance and capability building • Leads AI agenda, develops and implement AI strategy • Develop and enforce governance frameworks to ensure ethical and efficient AI use • Promote AI adoption and capability building • Enable entities through support and resources to initiate and scale their own AI projects
		A center of excellence offering a collaborative platform designed for government entities to innovate, test, and scale AI solutions aimed at addressing the challenges within the public sector • Accelerate the development of innovative AI solutions • Create a collaborative ecosystem for government entities to innovate and share knowledge and resources • Accelerate the deployment of AI technologies • Serve as an incubator for AI projects, providing technical and financial support for rapid prototyping and testing	An AI Hub aiming to bridge the gap between the public and private sectors, leveraging collective expertise, technology, and funding to foster AI-driven innovation and economic growth • Stimulate economic growth by nurturing AI startups and SMEs through funding, mentorship, and market access • Foster collaboration between the public and private sectors to co-develop AI solutions • Establish the emirate as an international hub for AI research, development, and innovation. • Attract local and international investments in AI technologies and enterprises

GovAI will lead the Steering, Adoption, and Governance of AI Initiatives Across the AD



GovAI

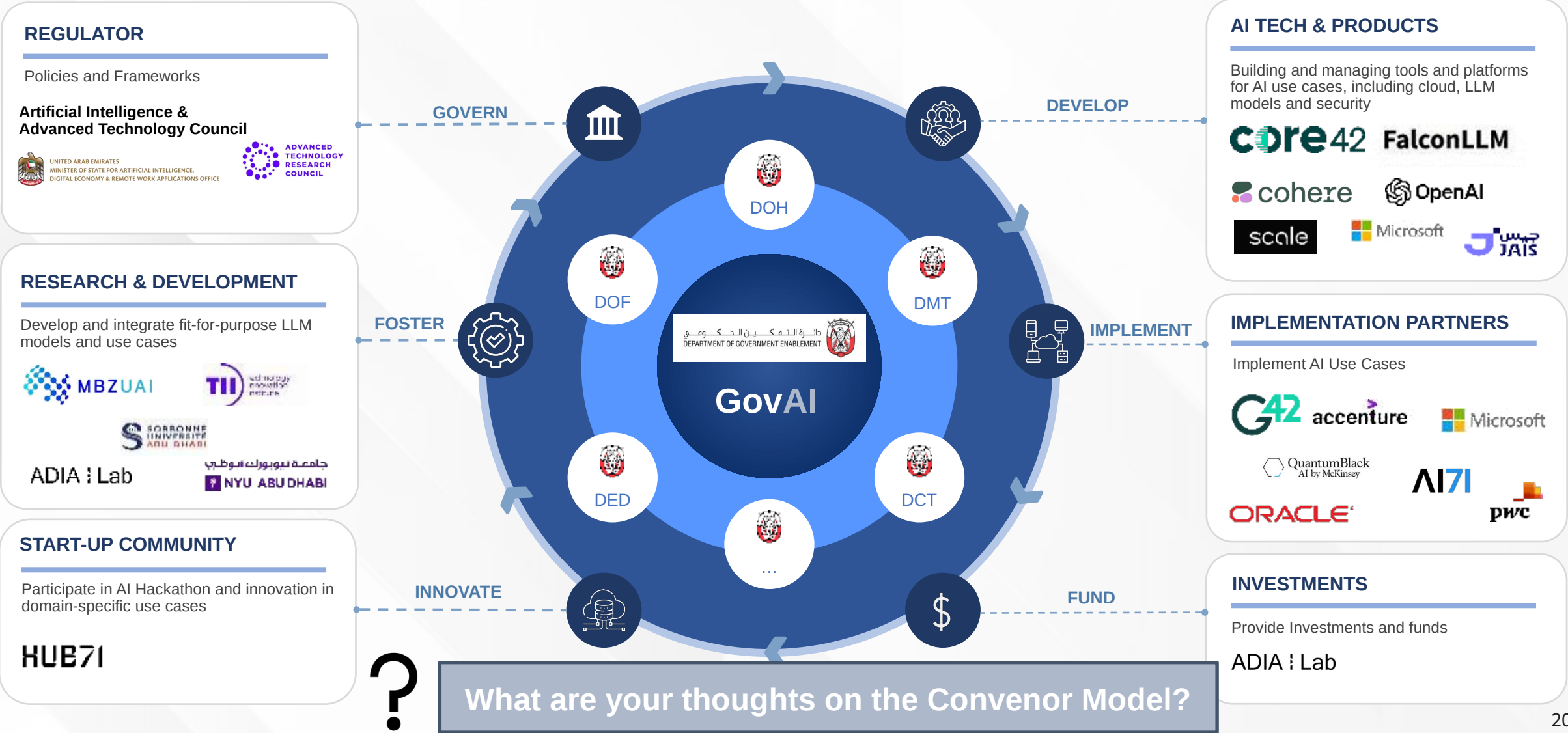
A **government-wide office** that act as a cornerstone for AI initiatives across the AD, tasked with **steering the AI agenda**, facilitating the **adoption of AI tech**, and ensuring **governance and capability building**

- **Leads AI agenda**, develops and implement AI strategy
- Develop and enforce **governance frameworks** to ensure ethical and efficient AI use
- Promote AI adoption and **capability building**
- **Enable entities** through support and resources to **initiate and scale** their own AI projects
- **Accelerate** the development of **innovative AI solutions**
- Create a **collaborative ecosystem** for government entities to **innovate** and share knowledge and resources
- **Accelerate** the deployment of **AI technologies**
- Serve as an **incubator for AI projects**, providing technical and financial support for rapid prototyping and testing



What are your thoughts on the scope of the office?

Achieving this ambition requires a **comprehensive ecosystem** including delivery, funding, tech and R&D partners



Our interaction with ADGEs across the lifecycle would be catered to their maturity

AI Use Case Lifecycle											
	Strategy & Governance	Use Case Identification and Detailing	Business Case Development	Use Case Approval	Enablement	Development	Responsible AI Compliance Check	Impact Validation	Scaling and Commercialization	Continuous Monitoring	
Step Owner / Accountability											
ADGE	✓ ADGE Framework	✓	✓	✓ ADGE Use Cases		✓ ADGE Use Cases			✓	✓ End-to-end (impact, tech, etc.)	
GovAI	✓ Whole-of-gov. Framework			✓ Whole-of-gov. Use Cases	✓	✓ Whole-of-gov. Use Cases	✓	✓		✓ Compliance, whole-of-gov impact	
GovAI Available Enablers*											
AI Strategy Co-creation		AI Training Curriculum	AI Training Curriculum	AI Use Cases Impact Validation		AI Funding Framework	AI Delivery Framework w/ Partners	AI Security & Ethical Certification	AI Value Assurance and Reporting	AI Funding Framework	AI Value Assurance and Reporting
AI Advisory pool of SMEs		AI Delivery Framework w/ Partners				AI Sandbox	AI Acceleration Funds			AI Acceleration Funds	
		Hackathons and Annual Competitions				Supercomputing	AI Training Curriculum			AI Delivery Framework w/ Partners	
						LLM Catalog					

*ADGEs can access enablers based on their maturity and needs



What are your thoughts on the roles and responsibilities?

Let's divide into 3 groups and explore some enablers

Group 1

Elaborate on what services are at the center vs the ADGEs

Group 2

Explore ideas for funding framework

Group 3

Explore how partnerships should be structured or engaged

Explore outcome-base contracts

Governments are tackling the AI talent hurdles to meet their AI vision



- **Thousand Talents Plan** in 2008 to recruit IT experts, primarily from overseas Chinese communities via dedicated university positions
- Specialized **AI and IT university programs**, e.g., Beijing Academy for AI



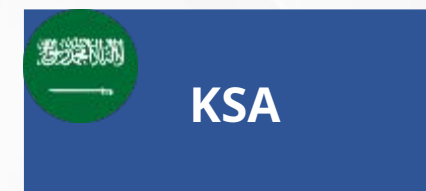
- **Investment of \$20 M** has been aimed at **tripling AI practitioners to 15,000 in 5 years**
- **\$7M AI Visiting Professorship and AI Accelerated Masters Program (AMP)**



- RISELab at **UC Berkeley** conducting **advanced AI research** focused on systems, security, and real-time intelligent execution to provide societal & industry solutions
- **Retention of elite students** in Bay Area to drive innovation in AI



- National AI strategy established **4 Interdisciplinary Institutes of Artificial Intelligence (3IA)** and targets **500 new PhDs in AI per year**
- Created 180 **additional academic chairs** and **300 additional PhDs in AI disciplines** at Notable universities like Ecole Polytechnique,



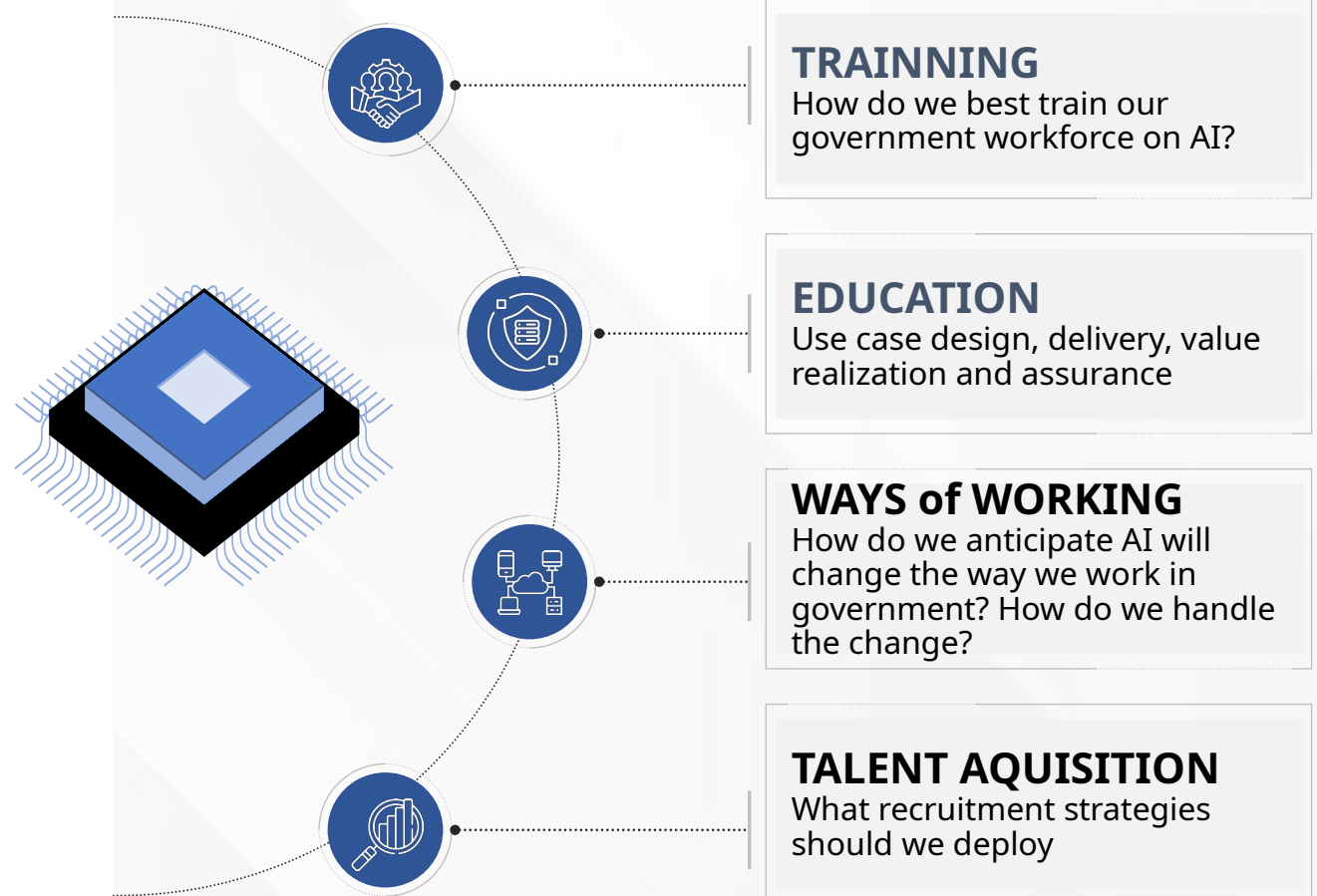
- Ministry of Communication and Information Technology launched an **AI training programme** to upskill engineering workforce and the **Artificial Intelligence Challenge** to encourage Saudi youth to drive innovation in AI

Talent is a key enabler, let's explore ways to address the challenges

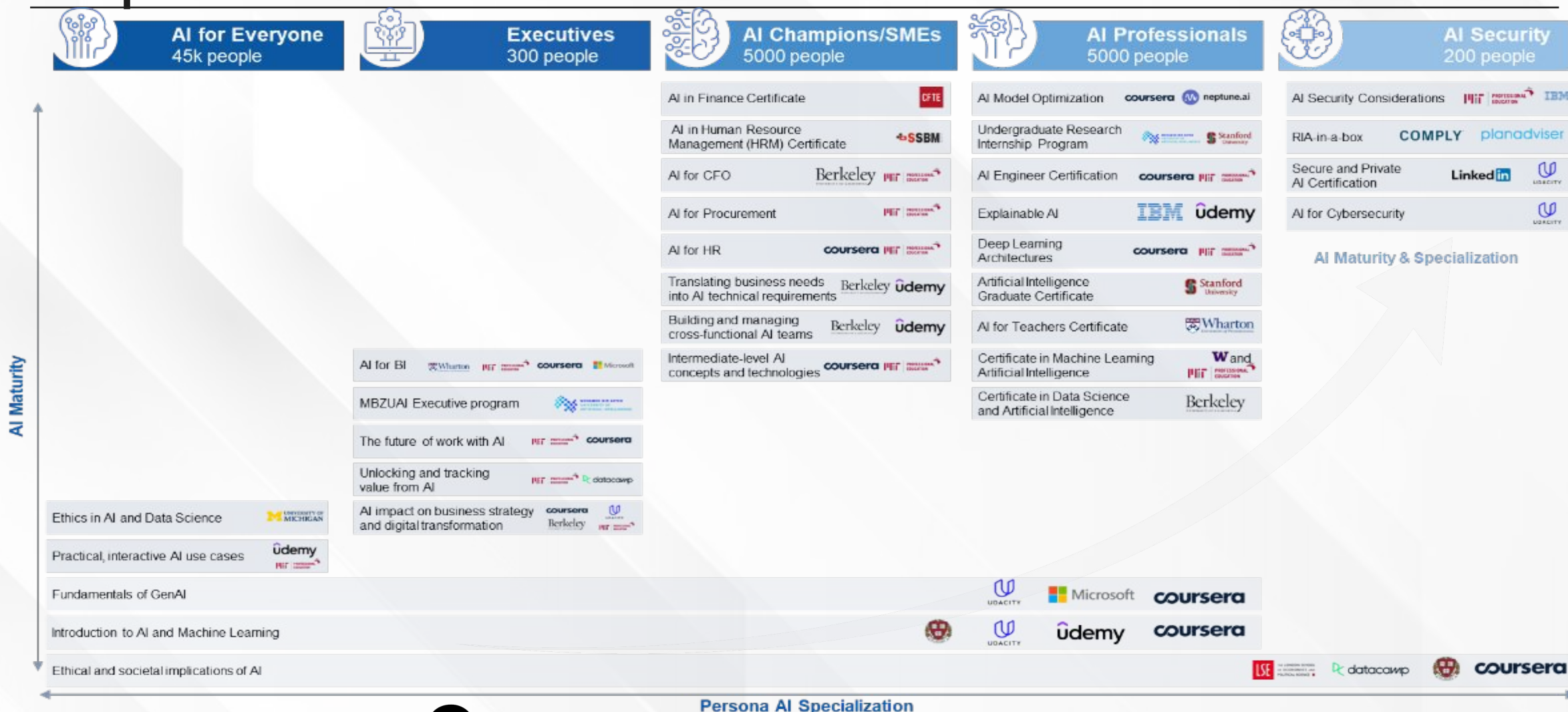
+45k of Government Employees
+300 Executives
+5k SMEs

+10k of University Students

Shaping AI organization vision, mission, and strategic objectives



A tailored AI Curriculum to upskill and augment Government employee competencies

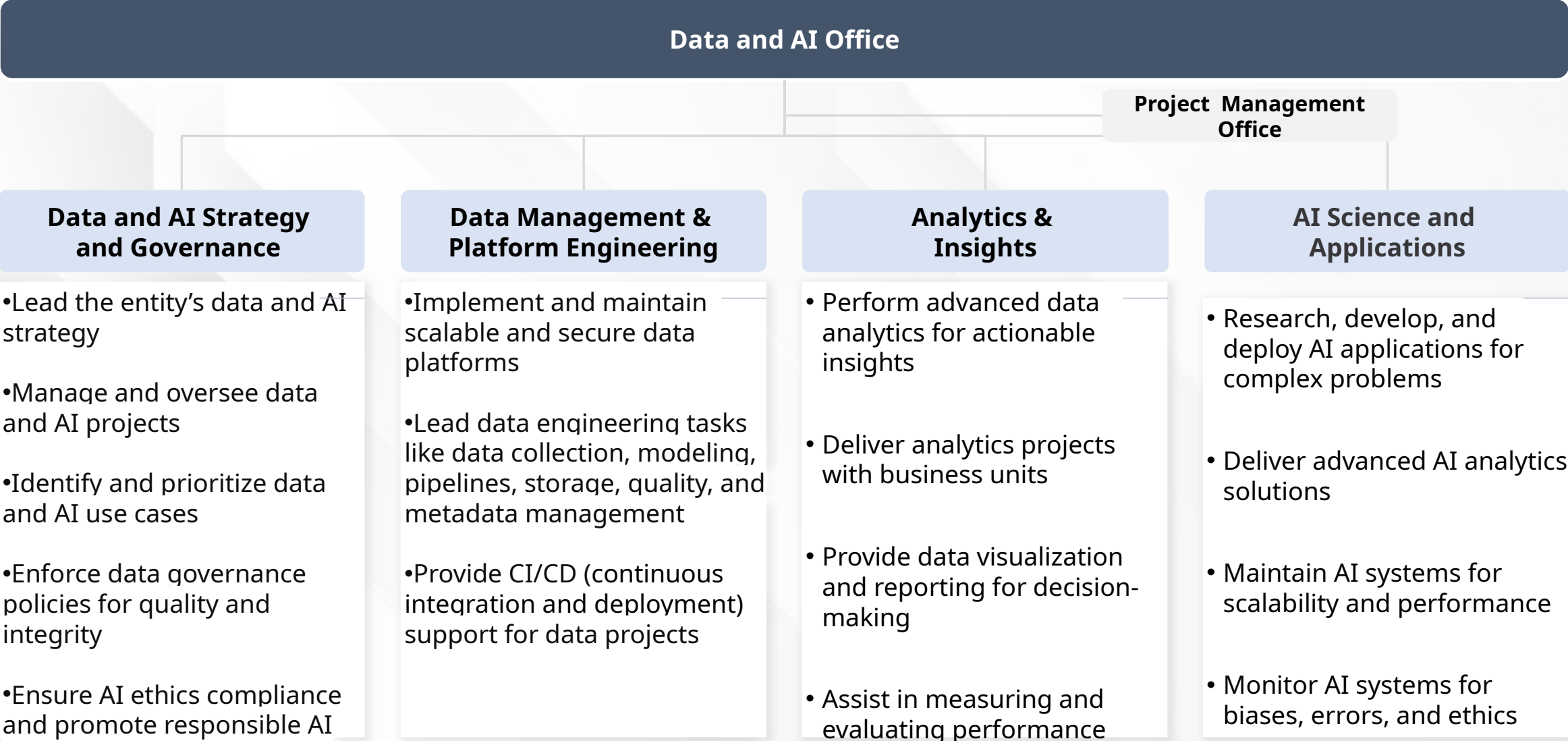


What are your thoughts?

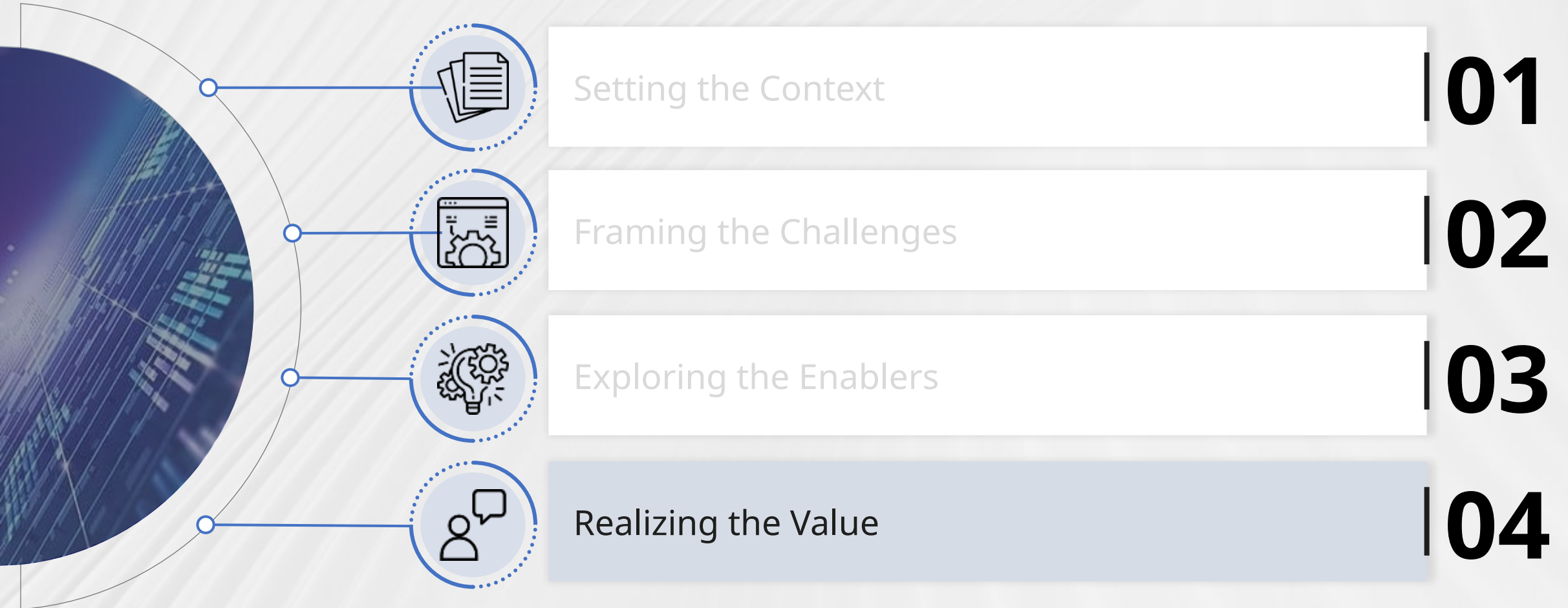
ADGEs Organizational Design Options for AI Adoption

	DEDICATED AI EXECUTIVE IN EACH DEPARTMENT	EVOLVING THE ROLE OF CHIEF DATA OFFICER	DEDICATED AI EXECUTIVE IN EACH DEPARTMENT
<u>Pros</u>	• Focused leadership • Tailored solutions	• Leverages existing roles • Cost efficient	• Centralized AI Governance with Departmental AI Leads
<u>Cons</u>	• Resource intensive • Coordination challenges	• Competing priorities • Variable expertise	• Complex structure • Potential conflicts

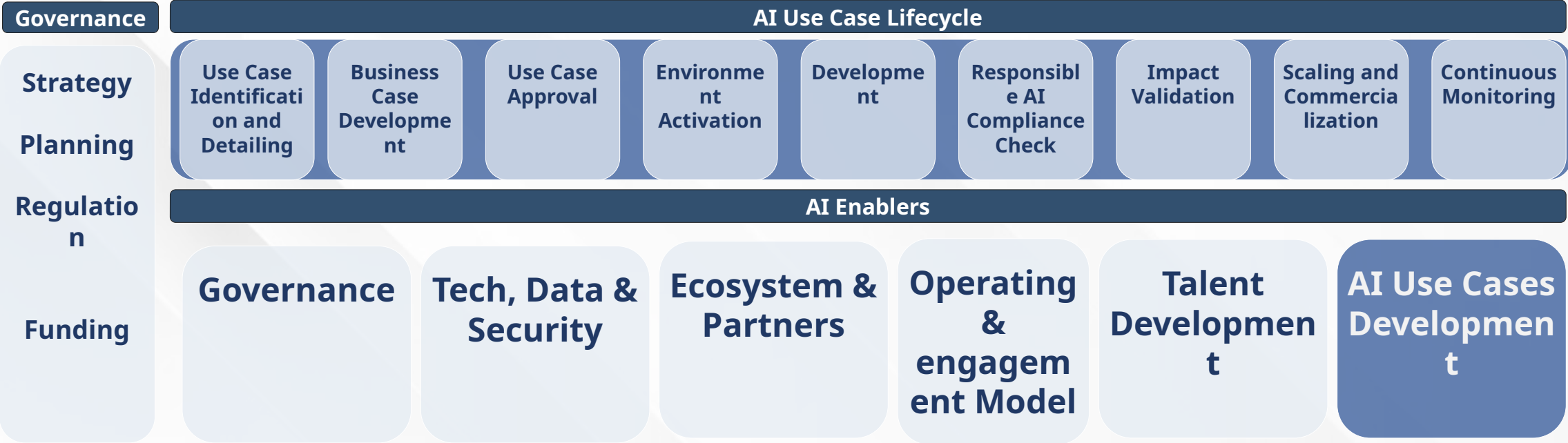
ADGEs will establish data and AI offices with 4 key capabilities



Agenda

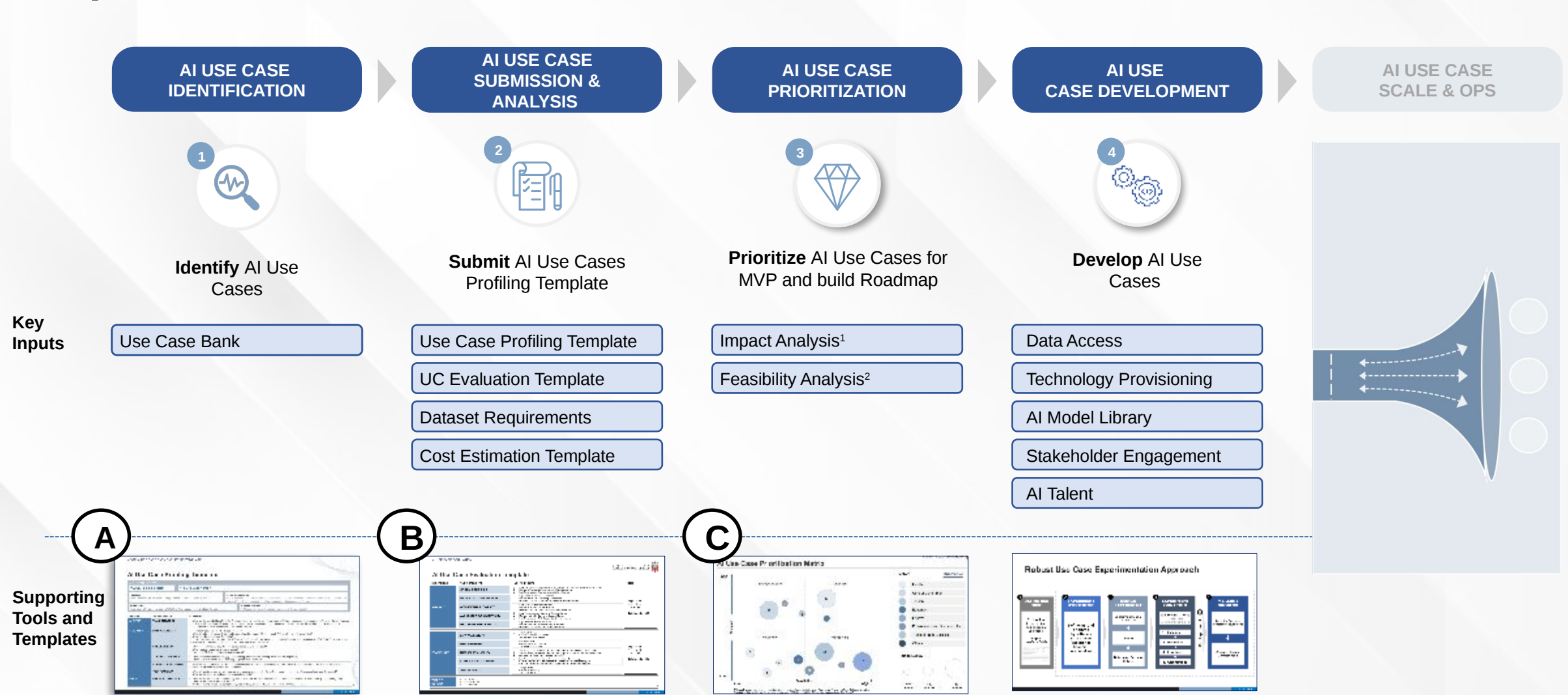


To address the challenges and achieve value from AI investments



Let's explore AI Use Case Life Cycle ...

We follow a standardized AI use case submission and prioritization process



1. **Expected Impact:** Citizen Experience, Productivity Enhancement, Operational Efficiency, Achievement of Gov. Goals and Output Quality
2. **Feasibility:** Data availability, Infrastructure readiness, AI Model availability, Stakeholder alignment and Cost feasibility

A

B

We defined 11 dimensions across Impact and Feasibility to assess AI use cases

AI Impact — Expected Value of the AI Use Case in Line with Government Priorities



Citizen Experience

Ability to improve quality and speed of user interaction with the government



Productivity Enhancement

Ability to increase process efficiency and cost sustainability through automation



Operational Efficiency

Ability of use case to improve worker output with new AI-powered applications and tools



Achievement of Govt. Goals

Extent to which use case enables achievement of horizontal and vertical goals through AI



Uplift Output Quality

Ability for government to deliver citizen and more mission outcomes effectively

Feasibility — Analysis of Complexity of AI Use Case Delivery as well as Estimated Cost and Time



Data Availability

Readiness of the variety and volume of data, and level of integration involved



Infra Readiness

Computational power, cloud services and infrastructure needed



AI Model Complexity

Complexity of the AI requirements, incl. AI algorithm, LLM or ML model needed



Stakeholder Alignment

Level of engagement with various internal and external stakeholders



Cost Feasibility

Estimated size of investment to develop and maintain the AI use cases

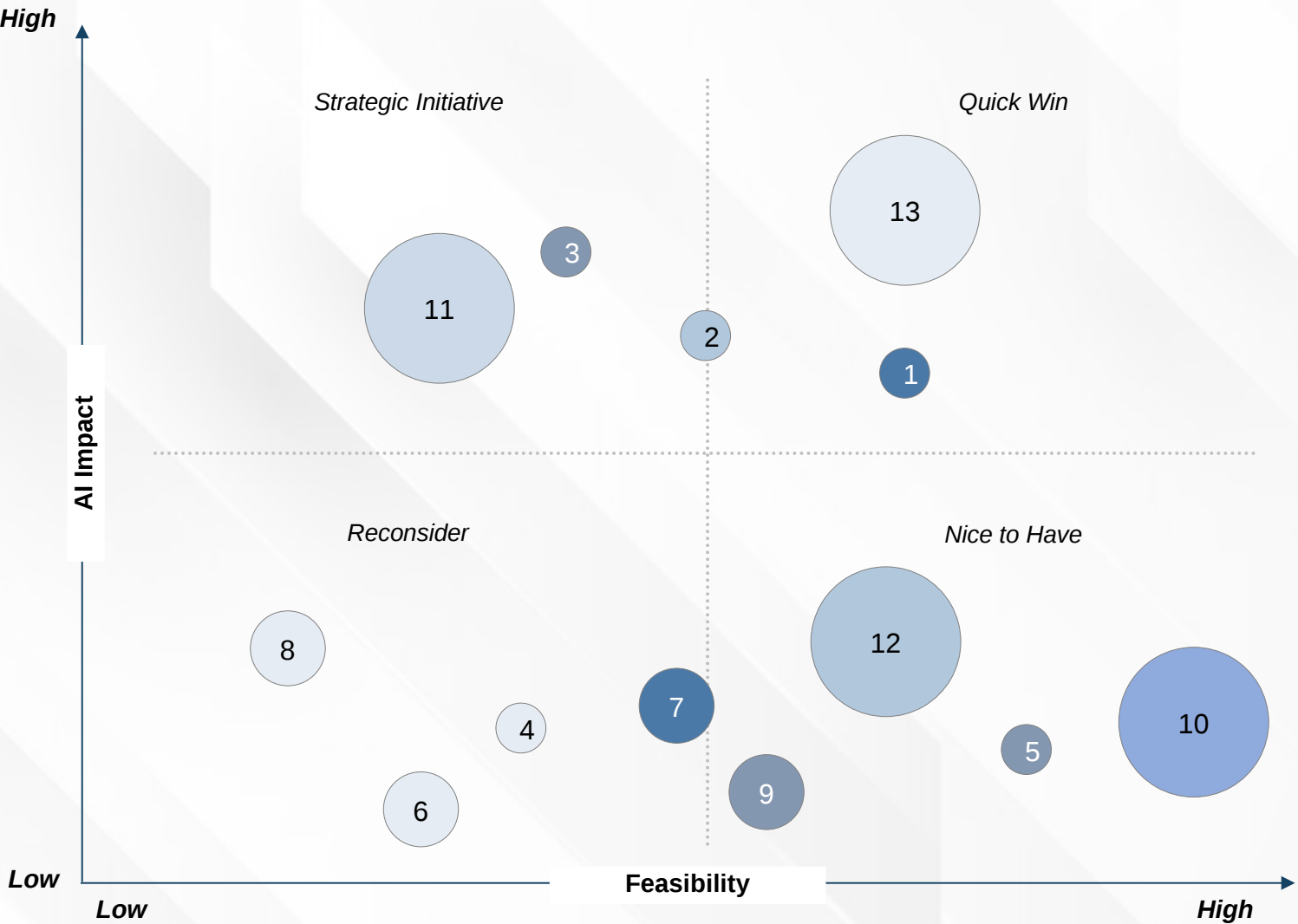


Time to Deploy

Estimated time to develop use case MVP

Time
MVP time to deploy

© AI Use Case Prioritization Matrix



Sectors

Health

Central Government

Tourism

Education

Energy

Environment and Sustainability

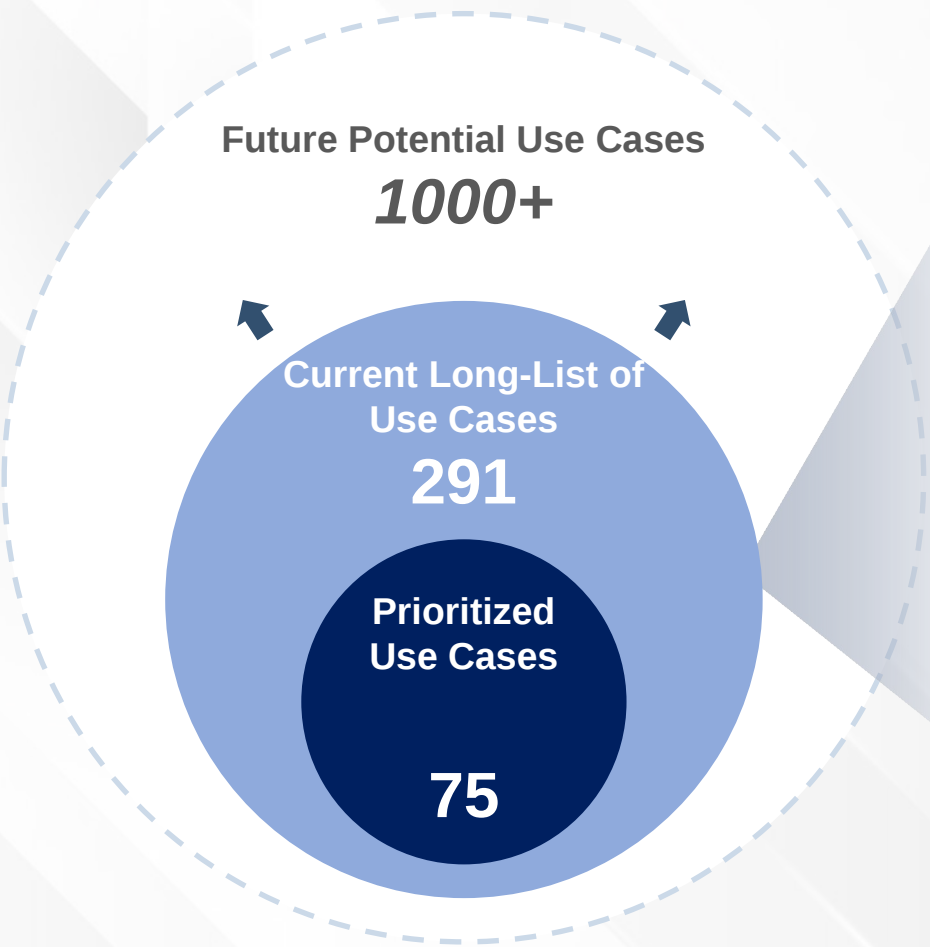
Transport and Logistics

Others

Time to Deploy0-6 months6-12 months12 months

This has resulted so far with 291 use cases collected and 75 prioritized

Abu Dhabi's AI Journey



ADGE Breakdown

	Long List	Prioritized
 ADAFSA	58	5
 ADEK	19	3
 ADP	23	3
 DCD	13	5
 DCT	20	5
 DED	24	5
 DGE	76	23
 DMT	19	9
 DOE	18	5
 DOH	8	7
 SCAD	13	5
	291	75

Note (*): Assessment is based on prioritized use cases only

10 Use Cases that would generate impact across Whole-of-Government

1 AI-Driven Business Licensing Automate and improve the end-to-end business licensing process for any type of business	2 Auto-Inspection & Compliance Automate the inspection process to improve overall compliance to sectoral standards and improve efficiency	3 Call Center AI-Agent AI-powered CC agent that would accurately interpret customer queries, recognize intents and streamline call routing and response	4 AI-powered Chatbots Conversational chatbot that would provide users with accurate guidance tailored to their specific needs	5 Corporate Services AI-Agent AI agents to support with improving efficiency of corporate functions (e.g., HR, Finance)
6 Domain Specific AI-Agent Specialized agents to support with improving knowledge ad process efficiency for sectors (e.g. health, education)	7 Marketing & Media Content Creation Generate tailored content for government marketing campaigns, social media outreach, and communication initiatives	8 AI-enabled GIS Enhancing GIS capabilities with AI across different layers to generate actionable information based on user requirements	9 AI Code Generator Using AI to develop code compliant with DevOps standards, streamlining the development process	10 Responsible AI Assessment Tool Using AI to assess and audit other AI models and outputs across dimensions (e.g. fairness, accuracy)

Let's divide into 3 groups to discuss the use cases

Group 1	Group 2	Group 3
Call Center AI-Agent	Auto-Inspection & Compliance	AI-Driven Business Licensing
Domain Specific AI-Agent	AI-powered Chatbots	Corporate Services AI-Agent
AI-enabled GIS	Marketing & Media Content Creation	AI Code Generator
Responsible AI Assessment Tool		



1

AI-Driven Business Licensing

MVP
Budget

~5M AED

Expected
Duration

6-12
Months

BUSINESS OVERVIEW

AI-driven business licensing refers to the integration various AI capabilities, such as machine learning, natural language processing (NLP), predictive analytics to streamline the processes for obtaining, renewing, and managing business licenses. These AI improvements include applications pre-filling, next-best action, verification of authenticity of records, risk assessment models, applicant guidance and support, regulations and compliance monitoring, insights for decision-making, and others

PAIN POINTS

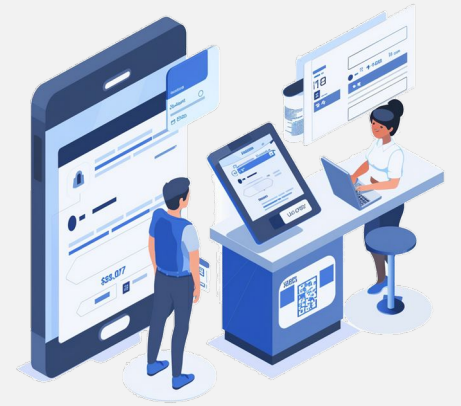
- Cumbersome and time-consuming licensing processes
- Friction in the investor journey
- Limited timely information to guide applicants/ investors
- Fraud detection and prevention
- Keeping up against changing regulations and increasing volumes

EXPECTED IMPACT

- Elevated Ease of Doing Business ranking
- Reduced business environment risks
- Streamlined investor journey
- Increase in business licenses
- Improved regulatory compliance
- Higher back-office productivity and lower costs

KEY RISKS

- Integrating AI systems with existing infrastructure and processes can be complex
- The use of AI in licensing may face regulatory hurdles
- Driving alignment among large number of licensing and approving entities
- AI algorithms could lead to unfair treatment of certain applicants or unwanted results



PRIORITY SCORING

IMPACT / VALUE

- 👤 Citizen Experience ☐
- 🔄 Productivity Enhancement ☐
- 📈 Operational Efficiency ☐
- 🏆 Achievement of Gov. Goals ☐
- 🏅 Uplift Output Quality ☐

FEASIBILITY

- 📊 Data Availability ☐
- 🏗️ Infrastructure Readiness ☐
- 🧠 AI Model Complexity ☐
- 👥 Stakeholder Alignment ☐
- 💰 Cost Feasibility ☐

2

Auto- Inspection & Compliance

MVP
Budget

~1M AED

Expected
Duration

0-6 Months

BUSINESS OVERVIEW

Traditional visual inspection today requires manual examination of products for defects and quality control, which is slow and error-prone. By integrating AI in the process of inspecting products, algorithms can automatically identify patterns, anomalies, or specific features that might be difficult to detect by human eyes. It works by aggregating data from multiple sensors into the AI model, enabling faster and consistent quality assurance. Continuously feeding the AI model with more data helps it improve faster and become more accurate over time.

PAIN POINTS

- Manual inspection of products and processes resulting in delays and backlogs in enforcement
- Resource constraints and scalability issues
- Inconsistencies in quality and thoroughness of inspections due to human error or oversight

EXPECTED IMPACT

- Enhanced public safety by enabling proactive and faster response time to potential risks and hazards
- Consistency of inspection quality, minimizing errors associated with manual inspections
- Improvement in productivity of inspection officers

KEY RISKS

- Integration issues with existing systems can be complex
- AI algorithms might perpetuate historical biases, resulting in discriminatory outcomes, which should be mitigated by integrating diverse training data into the model



PRIORITY SCORING

IMPACT / VALUE

	Citizen Experience	●
	Productivity Enhancement	●
	Operational Efficiency	●
	Achievement of Gov. Goals	◐
	Uplift Output Quality	●

FEASIBILITY

	Data Availability	◐
	Infrastructure Readiness	◐
	AI Model Complexity	◐
	Stakeholder Alignment	◐
	Cost Feasibility	◐

3

Call Center AI-Agent

MVP
Budget

~2M AED

Expected
Duration

0-6 Months

BUSINESS OVERVIEW

This use case revolves around augmenting call centers with AI agents to act as first line support for the call center incoming calls by seamlessly integrating with existing tools, managing incoming calls, and enhancing service offerings. It accurately interprets customer queries and recognizes their intents, streamlining call routing and information retrieval and provides tailored responses to individual preferences.

PAIN POINTS

- High volume of calls received (648K yearly)
- Difficulty in triaging inquiries to be handled effectively
- Timely access to call center support

EXPECTED IMPACT

- Call center agents can focus on critical asks, while common asks are handled by AI
- Reduced waiting time
- Reduced handling time
- Increase caller satisfaction

KEY RISKS

- Accuracy of STT, LLM and TTS engines (or multimodal LLM) to drive AI agent performance
- Ability to support Arabic effectively where 90% of the calls are on Emirati dialects
- Availability of up-to-date required knowledge from ADGE side



PRIORITY SCORING

IMPACT / VALUE

 Citizen Experience	●
 Productivity Enhancement	●
 Operational Efficiency	●
 Achievement of Gov. Goals	◐
 Uplift Output Quality	◐

FEASIBILITY

 Data Availability	●
 Infrastructure Readiness	●
 AI Model Complexity	◐
 Stakeholder Alignment	◐
 Cost Feasibility	◐

4

AI-powered Chatbots

MVP
Budget

~2M AED

Expected
Duration

0-6 Months

BUSINESS OVERVIEW

AI-powered chatbots are software applications that utilize AI to simulate human conversation. These chatbots can interact with users through text or voice interfaces, providing a more natural and intuitive user experience. They understand the user intent and request, analyze the user sentiment based on the provided inputs, and provide accordingly accurate recommendations directly or by routing the user to the right channel.

PAIN POINTS

- Overloaded call center agents
- Long wait times and queues through call center
- Variability in quality of support provided to users
- Generic responses provided by legacy chatbots

EXPECTED IMPACT

- Enhance user experience
- Reduce number of calls to agents
- Reduce waiting time and handling time
- Improve operational efficiency
- Improve agent productivity and resource allocation

KEY RISKS

- Accuracy and overall data quality of the current database such as standardized service names, descriptions and user information
- Ring-fencing AI chatbot responses within ethical / responsible boundaries
- Ability to support Arabic dialects effectively



PRIORITY SCORING

IMPACT / VALUE

 Citizen Experience	
 Productivity Enhancement	
 Operational Efficiency	
 Achievement of Gov. Goals	
 Uplift Output Quality	

FEASIBILITY

 Data Availability	
 Infrastructure Readiness	
 AI Model Complexity	
 Stakeholder Alignment	
 Cost Feasibility	

5

Corporate Services AI-Agent

MVP
Budget

~3M AED /
Corporate
Service

Expected
Duration

6-12
Months

BUSINESS OVERVIEW

The Corporate Services AI Agent enables every corporate function government employee to be assisted with a function-specific AI agent that understands the respective business areas (e.g. Human Capital, Finance, Procurement, etc.) and is trained on Abu Dhabi specific government data to advise on key decisions, support in operational processing, advise on demand requirements, provide recommendations based on best-practices, enable government employee proactive interventions, and others, in line with Abu Dhabi Government relevant regulations (e.g. Recruitment Standard, Salary Scale, Procurement Policies, etc.) and requirements.

PAIN POINTS

- Inefficiency in completing repetitive tasks
- Duplication of efforts
- Timely access to data and information
- Knowledge retention
- Potential for human error or inconsistency across different government employees

EXPECTED IMPACT

- Enhanced government employee efficiency
- Improved ability to comply to policies and standards
- Improved accuracy and consistency in completing tasks

KEY RISKS

- Access to sufficient and high-quality corporate service data to fine-tune AI models
- High resourcing needs to develop corporate services-specific LLMs
- Driving government employee adoption of corporate services-specific LLMs
- Integration with existing corporate service systems (e.g. ERP)



PRIORITY SCORING

IMPACT / VALUE

 Citizen Experience	
 Productivity Enhancement	
 Operational Efficiency	
 Achievement of Gov. Goals	
 Uplift Output Quality	

FEASIBILITY

 Data Availability	
 Infrastructure Readiness	
 AI Model Complexity	
 Stakeholder Alignment	
 Cost Feasibility	

6

Domain Specific AI-Agent

MVP
Budget

~5M AED /
Domain

Expected
Duration

>12
Months

BUSINESS OVERVIEW

The Domain Specific AI Agent enables every government employee to be assisted with a domain-specific AI agent that understands the industry (e.g. Health, Education, Customs, etc.) and is trained on Abu Dhabi specific government data to advise on key decisions, support in operational processing, present the latest industry trends, enable government employee proactive interventions, and others, in line with Abu Dhabi Government directives.

PAIN POINTS

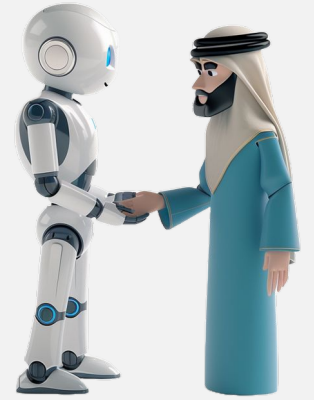
- Timely access to data and information
- Knowledge retention
- Potential for human error or inconsistency across different government employees

EXPECTED IMPACT

- Enhanced government employee efficiency
- Improved ability to achieve Government priorities (e.g. improved public safety, etc.)
- Improved accuracy and consistency in completing tasks

KEY RISKS

- Access to sufficient and high-quality domain data to fine-tune AI models
- High resourcing needs to develop domain-specific LLMs
- Driving government employee adoption of domain-specific AI agents



PRIORITY SCORING

IMPACT / VALUE

	Citizen Experience	●
	Productivity Enhancement	●
	Operational Efficiency	●
	Achievement of Gov. Goals	●
	Uplift Output Quality	●

FEASIBILITY

	Data Availability	◐
	Infrastructure Readiness	◐
	AI Model Complexity	○
	Stakeholder Alignment	◐
	Cost Feasibility	○

7

Marketing and Media Content Creation

MVP
Budget

~1.5M AED

Expected
Duration

0-6 Months

BUSINESS OVERVIEW

Integrating AI for Marketing and Media Content Creation automates the generation of targeted, high-quality content based on audience data and key trends. The AI model aggregates data from multiple sources, creating personalized and engaging media outputs. This streamlines the content creation process, ensuring timely and relevant communication with the public.

PAIN POINTS

- Time-consuming and costly content creation processes
- Difficulty in personalizing messages for diverse audiences
- Lack of cohesive messaging

EXPECTED IMPACT

- Higher public engagement tailored messages that resonate with diverse demographics
- Efficient communications, ensuring timely dissemination of information
- Optimizing resources allocation by reducing reliance on manual content creation

KEY RISKS

- Ensuring content quality and relevance
- Intellectual property management
- Oversight of the AI model to ensure performance is in line with ethical / responsible standards



PRIORITY SCORING

IMPACT / VALUE

 Citizen Experience	
 Productivity Enhancement	
 Operational Efficiency	
 Achievement of Gov. Goals	
 Uplift Output Quality	

FEASIBILITY

 Data Availability	
 Infrastructure Readiness	
 AI Model Complexity	
 Stakeholder Alignment	
 Cost Feasibility	

8

AI-enabled GIS

MVP
Budget

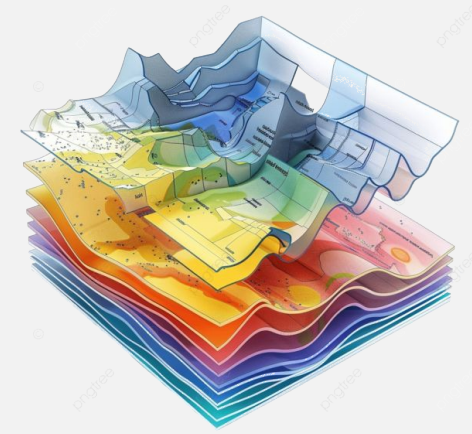
~5M AED

Expected
Duration6-12
Months

BUSINESS OVERVIEW

AI-enabled GIS refers to the integration of various GIS registers and embedding of AI technologies to enhance the analysis, interpretation, and management of spatial data. This leverages the strengths of AI, such as machine learning, computer vision, and predictive analytics, to improve the efficiency, accuracy, and capabilities of GIS.

The AI improvements include predictive modeling (urban growth, traffic patterns, or environmental changes), automated change detection, interactive 3-D maps for urban planning, real time of data processing for traffic and incident management, and others.



PAIN POINTS

- Limited ability to extract specific, actionable insights from existing GIS maps
- Keeping GIS data and systems up-to-date and well-maintained
- Time-consuming manual searches and analysis
- Data accessibility and usability

EXPECTED IMPACT

- Enhanced decision-making for urban planning and public services
- Improved infrastructure management
- Improved environmental conservation
- Optimized resource management
- Well-maintained, more reliable and accessible spatial data

KEY RISKS

- Integration challenges with existing GIS systems
- Consistency of quality of maps / images
- Regulatory hurdles for adopting AI in GIS applications

PRIORITY SCORING

IMPACT / VALUE

	Citizen Experience	●
	Productivity Enhancement	●
	Operational Efficiency	●
	Achievement of Gov. Goals	◐
	Uplift Output Quality	●

FEASIBILITY

	Data Availability	●
	Infrastructure Readiness	●
	AI Model Complexity	◐
	Stakeholder Alignment	◐
	Cost Feasibility	◐

9

AI Code Generator

MVP
Budget

~2M AED

Expected
Duration

0 – 6
Months

BUSINESS OVERVIEW

The AI code generator leverages AI to assist IT departments and software developments more specifically in writing, optimizing, and managing code in line with leading standards. It is trained on large volumes of code and related documentation in different coding languages.

PAIN POINTS

- Time-consuming coding processes
- Difficulty in maintaining consistent coding standards

EXPECTED IMPACT

- Faster time to code / development cycles
- Reduction in technical errors
- Higher adherence to leading standards
- Improved code quality

KEY RISKS

- Ensuring the accurate interpretation of coding requirements
- Ensuring the accuracy and reliability of AI-generated code
- Potential for over-reliance on automated code generation without proper oversight



PRIORITY SCORING

IMPACT / VALUE

 Citizen Experience	
 Productivity Enhancement	
 Operational Efficiency	
 Achievement of Gov. Goals	
 Uplift Output Quality	

FEASIBILITY

 Data Availability	
 Infrastructure Readiness	
 AI Model Complexity	
 Stakeholder Alignment	
 Cost Feasibility	

10

Responsible AI Assessment Tool

MVP
Budget

~2.5M AED

Expected
Duration0 – 6
Months

BUSINESS OVERVIEW

The Responsible AI Assessment tool evaluated the quality and compliance of AI-based products against different responsible AI dimensions (including fairness, explainability, accuracy, and other relevant dimensions). The tool assesses AI application risk by testing the model, and provides a consolidated pass/fail report broken down by dimension for compliance approval, ensuring readiness of the AI product for deployment and scaling.

PAIN POINTS

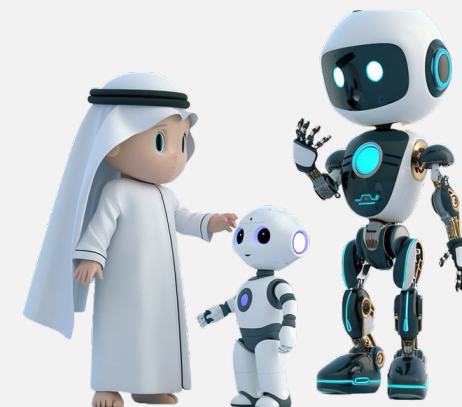
- Limited trust of business users in some AI models
- Challenges in assessing AI model risks and understanding its reliability
- No readily available tool to qualify AI model meets ethical and regulatory compliance criteria before operationalization

EXPECTED IMPACT

- Improved trust and reliability in AI systems
- Avoidance and management of AI risks
- Enhanced regulatory compliance and ethical standards

KEY RISKS

- Ensuring the robustness and accuracy of the responsible AI assessment tool
- Developing a tool that can manage a wide variety of AI model types
- Embedding the tool seamlessly into the AI development workflow



PRIORITY SCORING

IMPACT / VALUE

 Citizen Experience	☒
 Productivity Enhancement	☒
 Operational Efficiency	☒
 Achievement of Gov. Goals	☒
 Uplift Output Quality	☒

FEASIBILITY

 Data Availability	☐
 Infrastructure Readiness	☐
 AI Model Complexity	☐
 Stakeholder Alignment	☐
 Cost Feasibility	☐

TAMM 3.0 powered by AI

Khalifa wants to renew his vehicle

To achieve this, he must undertake the following steps



01

**Vehicle
inspection**



02

**Select insurance type
and purchase it**



03

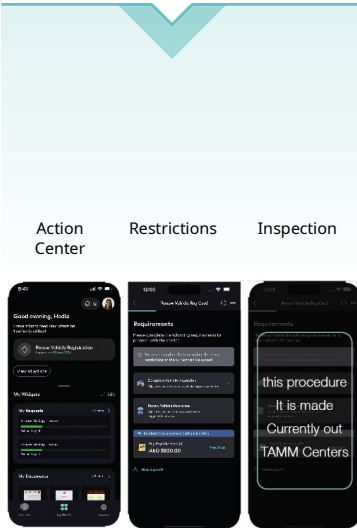
**Pay registration fees
and traffic fines**



Current Service Flow
Vehicle registration Journey – 20 Steps

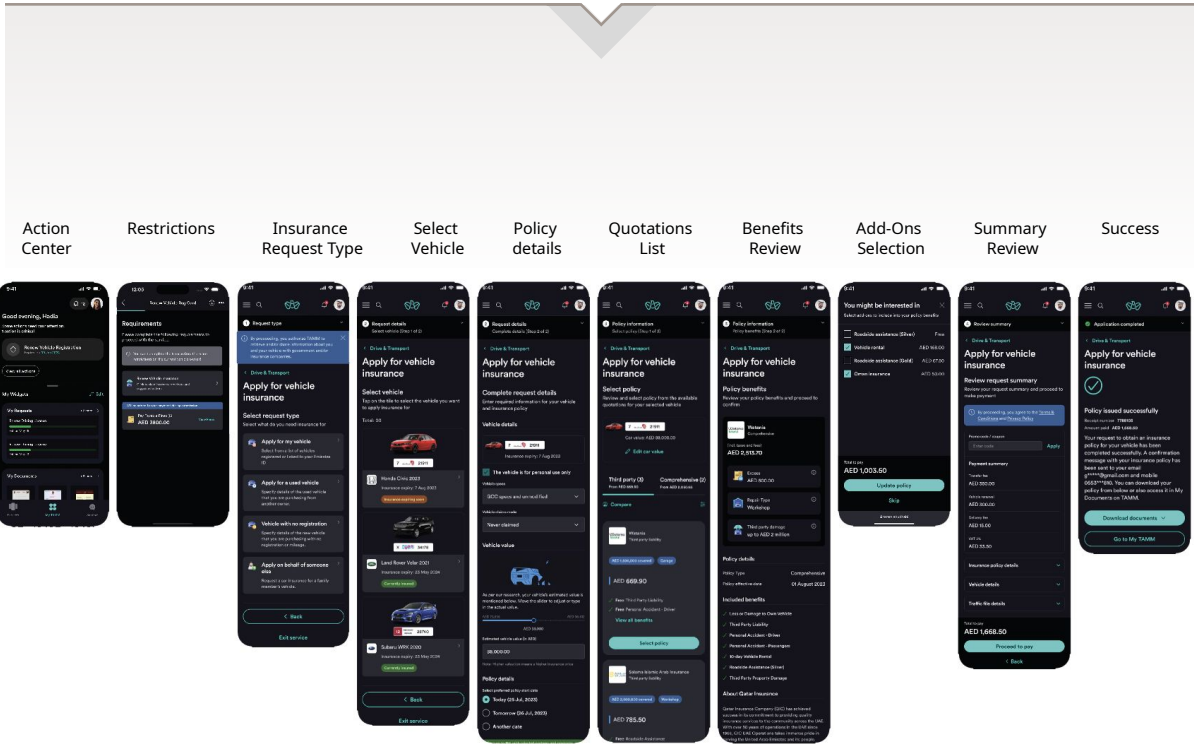
VEHICLE
INSPECTION

Steps 3



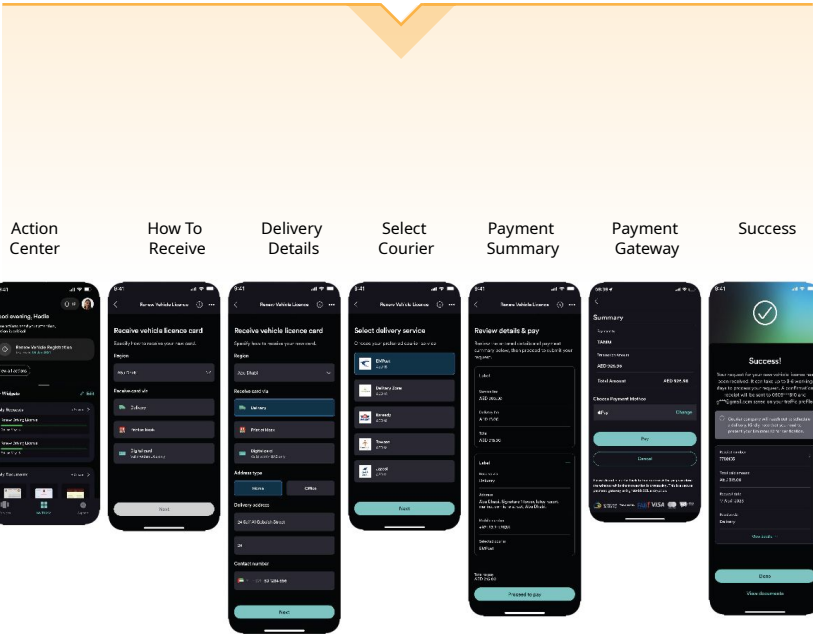
VEHICLE
INSURANCE

Steps 10



FINE PAYMENT &
VEHICLE LICENCE RENEWAL

Steps 7





Current verses future services

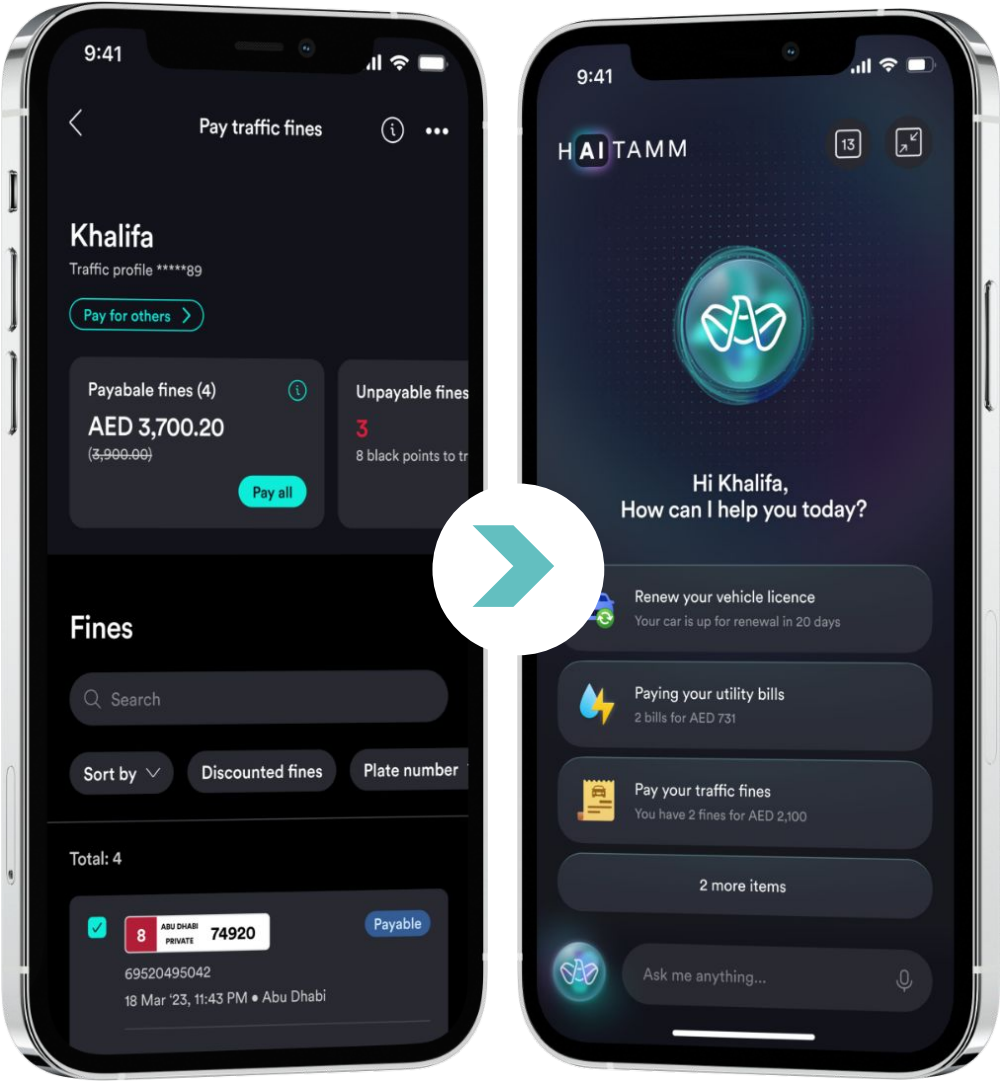
Current service flow

20

Number of screens to finalize the services

19 Minutes

Estimated time to finalize the service



Future Service flow

2

Number of conversations to finalize the service

5 Minutes

Estimated time to finalize the service

























AI “Big Ideas”

These ideas aim to capture the world's imagination, generate widespread media coverage, and firmly establish Abu Dhabi as the AI Government global leader. By pushing the boundaries of what's possible with AI and showcasing its transformative potential in bold and exciting ways, we can attract top talent, accelerate innovation, and create a truly visionary future for Abu Dhabi



Abu Dhabi AI Hackathon

Partnering with OpenAI, host an annual global AI challenge series covering themes like healthcare and education. Offering substantial rewards for solutions applicable in Abu Dhabi, these competitions attract global experts and nurture local talent, building a strong AI community.



“AI for You” Campaign

Launch an 'AI for You' campaign to boost AI literacy citywide. Featuring free online courses, public lectures, interactive exhibitions, and a mobile lab visiting schools and community centers, it aims to spark enthusiasm and understanding of AI, prompting citizens to explore AI's potential in their professions and daily lives.



X-Prize for AI Innovation

Propose an Abu Dhabi X-Prize for government AI Use Cases, offering a significant prize to global innovators tackling city or regional challenges with AI solutions. Areas of focus could range from climate change to healthcare and urban planning. This initiative positions Abu Dhabi as a leading AI hub, drawing top talents and thinkers worldwide.



AI Olympiad Talent Competition

Launch an annual 'AI Olympiad,' inviting global talent to tackle real-world challenges in Abu Dhabi. With substantial prizes and government partnership opportunities, this initiative highlights Abu Dhabi's commitment to AI for social good.



Holographic Government Experience

Introduce the pioneering 'Holographic Government,' enabling citizens to interact with lifelike holograms of officials and service providers. This immersive experience facilitates access to services, virtual town halls, and engaging with leaders, positioning Abu Dhabi as a leader in cutting-edge citizen engagement.

Abu Dhabi's ambition is to become an AI Native Government achieving Whole-Of-Emirate Impact



Improve the quality and speed of citizen & resident interaction with the government



Increase process efficiency through automation to become more cost efficient



Improve worker output with new AI-powered applications and tools



Enable achievement of horizontal and vertical Government goals through AI



Uplift Government output quality by improving accuracy and citizen engagement

دائرة التمكين الحكومي
DEPARTMENT OF GOVERNMENT ENABLEMENT



Thank You

May 2024



Appendix

May 2024

PRIORITIZED ADGES AI USE CASES

ADGEs prioritized AI use cases

ADGES	(A) H2 2024 (0-6 months)	(B) H1 2025 (6-12 months)	H2 2025 (>12 months)	Total
 ADAFSA	1	4	-	5
 ADEK	1	2	-	3
 ADP	1	1	1	3
 DCD	-	5	-	5
 DCT	1	2	2	5
 DED	-	3	2	5
 DGE	14	8	1	23
 DMT	4	4	1	9
 DOE	-	1	4	5
 DOH	5	2	-	7
 SCAD	-	4	1	5
Total	27	36	12	75

A ADGE PRIORITIZED USE CASES 0-6 MONTHS (1/3)

ADGE Name	Use Case Title	Use Case Description	AI Impact	Feasibility	Time to Deploy
★ ADAFSA	AI Food Safety inspection parameters detected (Expansion Phase)	The purpose of this use case is to replace manual food safety monitoring with AI-driven cameras, enhancing compliance and safety practices by automatically detecting violations and continuously monitoring food establishments. This project aims to improve food safety standards, reduce human error, and ensure timely interventions to prevent foodborne illnesses.	●	●	0-6 Months
ADEK	Education Insights	The purpose of this use case is to enhance the educational system managed by leveraging AI to analyze data from PISA and other international assessments (being in the Emirate's long-term agenda). This aims to deepen insights, elevate knowledge levels, and expand data analysis capabilities	●	●	0-6 Months
ADP	Mini Bus Violation System	Minibuses are an anomaly on current r installed radar systems, due to being categorize as light vehicle. These buses are utilized to transport passengers (Capacity of 14 max), however due to previously motioned challenge the drivers usually violated buses speed limited which endanger the lives of these passengers. Utilizing Computer vision Models allows to violate minibuses that are going over the speed limit	●	●	0-6 Months
DCT	AI Based Literature Classification	The purpose of this use case is to deploy an AI-driven model to help Emirati and Arab publishers classify literature based on readability levels, ensuring accurate targeting of readers across various age groups.	●	●	0-6 Months
DGE - GovAcademy	Individual Development Plan	The purpose of the use case is to improve individual development by mapping job profiles and facilitating professional growth and career advancement	●	●	0-6 Months
DGE - GovAcademy	Training Advisor	The purpose of the use case is to provide guidance and support to employees in identifying, accessing, and navigating training and development opportunities	●	●	0-6 Months
DGE - Procurement	AI enabled Spend Analytics	The Purpose of the use case is to alleviate the problem of complex and time-consuming activity of categorizing voluminous spend data into the defined category taxonomy (UNSPSC)	●	●	0-6 Months
DGE - GovTalent	Labor Market Forecasting	The purpose of the use case is to utilize advanced predictive analytics to anticipate future job demands, the reason behind that is to try to enable the development of targeted hiring strategies and policy directives to align with emerging Abu Dhabi Government workforce (white color) needs	●	●	0-6 Months
DGE - GovTalent	Automated Job Matching	The purpose of the use case is to utilize automated algorithms to swiftly analyze resumes and pinpoint the best-fit candidates, the reason behind that is to try to significantly streamline the initial screening process and minimize time spent on candidate evaluation	●	●	0-6 Months
DGE - GovTalent	Interview Support	The purpose of the use case is to leverage AI to automatically review CVs, the reason behind that is to assess suitability and generate relevant interview questions based on the candidate's background and job requirements	●	●	0-6 Months
DGE - GovTalent	Crafting Job Descriptions for Posting	The purpose of the use case is to utilize AI to generate job descriptions, the reason behind that is to optimize diversity and search visibility and ensure job descriptions attract a broad range of qualified candidates	●	●	0-6 Months
DGE - TAMM	Contact Center Voice Bot	The voice bot leverages speech to text, NLP, text to speech to act as first line support for the CC incoming calls by seamlessly integrating with existing tools, managing incoming calls, and enhancing service offerings	●	●	0-6 Months
★ DGE - TAMM	Automated Case Triage	The purpose of the use case is to deploy automated case triaging, the reason behind that is to analyze incoming support tickets and automatically classify and prioritize them based on priority, type of incident, topic and channel the incident to the right stakeholders	●	●	0-6 Months

A ADGE PRIORITIZED USE CASES 0-6 MONTHS (2/3)

ADGE Name	Use Case Title	Use Case Description	AI Impact	Feasibility	Time to Deploy
DGE - TAMM	Knowledge Base Content	The purpose of the use case is to turn customer enquiries and incidents resolution into clear, comprehensive and easy-to-read knowledge articles by sorting, labeling and transforming them in English and Arabic, classifying and categorizing them as per the existing KA forms and leveraging different intelligent and search tools to present them to both customers and agents			0-6 Months
DGE - TAMM	Tailor knowledge and training solutions for individual frontline agents	The purpose of this use case is to deploy AI technologies to deliver personalized knowledge and training solutions to frontline agents in customer service or support roles, the reason behind that is to enhance the performance, efficiency, and effectiveness of individual agents by providing them with targeted learning resources and on-demand support The use case is aimed towards frontline agents to help them better leverage existing knowledge (explicit and tacit) across the organization and optimize their productivity			0-6 Months
DGE - TAMM	Identify the effectiveness of knowledge articles by linking it to outcome / customer satisfaction	The purpose of this use case is to evaluate the effectiveness of the defined and availed knowledge articles for the purpose of improving customer satisfaction, knowledge sharing effectiveness and useability, by measuring their impact on outcomes The key beneficiaries of this use case are primarily the contact center agents and beneficiaries			0-6 Months
DGE - TAMM	Voice-to-Text Transcription	The purpose of this use case is to transcribe voice interactions between customers and call center agents into text in real-time, enabling faster and more accurate documentation of support cases. This use case primarily benefits contact center agents and end-users by streamlining the support process			0-6 Months
★ DGE - CX	AI-powered Conversational Chatbots	The purpose of the AI-powered Conversational Chatbots is to integrate extensive customer experience service catalogue encompassing detailed information such as service names, descriptions, application procedures, associated costs, and required documentation into AI chatbots, to provide users with accurate guidance tailored to their specific service needs It mainly benefits end-users and agents			0-6 Months
DMT	Construction Approval, Development & Monitoring	The purpose of this use case is to employ AI-based monitoring systems for quality control and project oversight. Leveraging AI's diagnostic capabilities, they identify root causes of deviations, enabling proactive measures to optimize project outcomes and mitigate risks effectively Key beneficiaries for this use case include Citizens, Abu Dhabi Leadership, DMT Leadership and ADREC			0-6 Months
DMT	Sentiment Analysis based on Services Comments	The purpose of the use case is to use text data to understand public opinions, identify whether feedback is positive, negative, or neutral. The use case also helps decision-makers improve services based on public sentiment Key beneficiaries of this use case include Citizens, DMT and Affiliates			0-6 Months
DMT	Traffic Management	The purpose of this use case is to leverage data analytics and real-time monitoring to optimize traffic flow, reduce congestion, and enhance road safety. It seeks to integrate various data sources such as traffic cameras, vehicle sensors, and weather forecasts to improve the efficiency of transportation networks and the overall commuting experience Key beneficiaries for this use case include the whole of Abu Dhabi society such as residents and commuters and public safety operators			0-6 Months

A

ADGE PRIORITIZED USE CASES 0-6 MONTHS (3/3)

ADGE Name	Use Case Title	Use Case Description	AI Impact	Feasibility	Time to Deploy
DMT	Real State Market Prediction	The purpose of the use case is to leverage data analysis and machine learning to forecast trends, pricing changes, and demand fluctuations in the real estate market			0-6 Months
DOH	Patient Risk Profile	The purpose of this use case is to identify the probability a patient has of developing a disease (14 diseases e.g., diabetes, hypertension) and provide the risk level to physicians for more accurate and timely diagnosis, for overall improved patient outcomes			0-6 Months
DOH	Chat with your Data	The purpose of this use case is to provide real time answers to questions on all available healthcare data (e.g., disease registries, etc.) making data easier and quicker to access and interpret, and allowing easier access to insights and indicators			0-6 Months
DOH	Personalized Behavioral Interventions	The purpose of this use case is to provide behavioral nudges for the population, leveraging AI, to increase their physical activity levels with the aim to reduce overall obesity rates			0-6 Months
DOH	Anomaly Detection	The purpose of this use case is to proactively detect and flag anomalous physician and provider transactions (claims, sick leaves and prescription of select drugs) initiating ongoing investigations for improvements in outcomes and experiences			0-6 Months
DOH	AI Symptom Checker	The purpose of this use case is to provide an AI-based symptom diagnostic tool based on symptom-to-disease correlation. Such a tool enables patients to determine potential diagnoses and risk levels for their potential conditions.			0-6 Months

B ADGE PRIORITIZED USE CASES 6-12 MONTHS & >12 MONTHS (1/5)

ADGE Name	Use Case Title	Use Case Description	AI Impact	Feasibility	Time to Deploy
ADAFSA	Abu Dhabi Agriculture & Food Safety Data Platform Phase 2	The purpose of this use case is to provide leadership dashboards and predictive insights for food security and the agricultural sector, integrating various sources of agricultural data. This platform supports enhanced decision-making by providing up-to-date information on global price indices, supply chain threats, food safety data, and more.			6-12 Months
★ ADAFSA	AI Agricultural inspection	The purpose of this use case is to utilize aerial photos, farm-related data, and AI to monitor farms, assess compliance, detect violations, and identify areas needing attention, thereby optimizing inspections and monitoring activities in the agricultural control and extension sectors.			6-12 Months
★ ADAFSA	AI Veterinary	This use case aims to support both doctors and livestock owners in managing the well-being of livestock through an AI-powered diagnostic, reporting, and treatment planning platform. It facilitates visual preliminary diagnoses, test result notifications, outbreak alerts, and remote veterinary consultations.			6-12 Months
ADAFSA	AI Import Diversification Engine	The purpose of this use case is to monitor global prices, supply chain disruptions, natural disasters, and macroeconomic trends to recommend optimal trade partners for food products. The system aims to optimize costs, diversify suppliers, and manage risks by providing real-time data and predictive insights.			6-12 Months
ADEK	AI to create Intelligent Tutoring Systems (ITS)	The purpose of this use case is to use AI to create Intelligent Tutoring Systems (ITS) that provides personalized instruction and feedback for students, thereby providing individualized attention to each student and overcoming the challenge of the student-teacher ratio. This would enhance the student experience, alleviate teachers' workload, and keep parents well informed of their children's progress			6-12 Months
ADEK	Customized Pathways	The purpose of this use case is to enhance the educational offerings by providing customized pathways for academics, physical activities, or artistic talents. These will allow students to pursue individual interests and strengths, leading to improved engagement and outcomes. However, implementing these pathways requires careful consideration of resource allocation and flexibility. Understanding a school's capacity to offer these personalized options is crucial for effective planning and execution			6-12 Months
ADP	Phone and Seat Belt Violations	One of the main challenges we face in road's are road accidents that are usually caused by the driver being distracted by focusing on his phone. Even with improvement in road infostructure and educational Campaigns to enlighten drivers, still most of accidents are caused from human error. Utilizing Computer vision Models, we could utilize it to detect drivers who are using their phones, and divers with passengers who not wearing their seatbelt to issue violations in order to lower the accidents in the road.			>12 Months
ADP	Fines Auto Processing	Before a fine is issued a human operator validates and fill the vehicle details before issuing it and fining the vehicle. The challenge is the huge queue of fines that are waiting to be procced by the human operator. This inherently increase the duration between the fine registering from the radar to be issued out. With the advancements in computer vision and ANPR technology an ensemble models of ANPR models can Automate the fine issuing stage with significantly lower error rates with human operators, and slashes the duration of fines issuing significantly			6-12 Months
DCD	Family Planning Advisory	The purpose of the use case is to provide personalized and accessible guidance and support to individuals and families to help individuals make informed decisions about family planning and achieve their desired reproductive goals			6-12 Months
DCD	Detecting District Dysfunctionality	The purpose of the use case is to detect signs of dysfunctionality within districts of the Emirate of Abu Dhabi. The system aims to identify potential areas of concerns within the districts in real-time, allowing timely intervention and targeted allocation of resources to address emerging issues			6-12 Months
			 Low	 Medium	 High

B ADGE PRIORITIZED USE CASES 6-12 MONTHS & >12 MONTHS (2/5)

ADGE Name	Use Case Title	Use Case Description	AI Impact	Feasibility	Time to Deploy
DCD	Identify Vulnerable Citizens	The purpose of the use case is to identify and assist vulnerable individuals within a community. The use case aims to identify individuals who may be at risk due to factors such as poverty, health issues, disabilities, or social isolation.			6-12 Months
DCD	AI-Enhanced Community Engagement	The Purpose of the use case is to enhance community engagement for both donations and volunteering			6-12 Months
DCD	Detect Likelihood of Relapse	The Purpose of the use case is to assess the likelihood of relapse in substance abusers			6-12 Months
★ DCT	Marketing Media Buying Optimization	The purpose of the use case is to deploy an AI model designed to optimize budget allocation efficiently across various source markets and digital media platforms for marketing campaigns			6-12 Months
DCT	DCT Corporate Insights GPT	The purpose of this use case is to address the challenge of manual report generation and KPI analysis by automating these processes through Gen AI. It offers users the opportunity to streamline data-driven decision-making, saving time and resources while enabling insightful analysis without specialized expertise			>12 Months
DCT	DCT Culture Insights GPT	The purpose of this use case is to transform art and culture appreciation through utilizing GenAI to create digital personas of artists, enabling museum visitors to interact via a chatbot for in-depth discussions about artworks, creative processes and artists' backgrounds, all initiated through QR codes.			>12 Months
DCT	AD Tourism Spatial Temporal Forecasting	The purpose of this use case is to deploy an AI based model to forecast hotel operational metrics, including guest count, number of nights, occupancy rates and revenue, to enhance operational and financial planning.			6-12 Months
DED	Real Disposable Income Analysis	The purpose of the use case is to leverage AI for real-time income tracking using various data sources (e.g., bank transactions) for a view on disposable income across different demographics and using predictive AI to forecast how disposable income might change based on economic indicators, policy changes, and other factors.			6-12 Months
DED	Predict Product market fit for Non Oil Exports	The purpose of the use case is to simulate scenarios, test policy changes, and ultimately make data-driven decisions for resource allocation, infrastructure development, and trade agreements, fostering economic diversification and growth beyond oil			>12 Months
DED	AI for Identifying potential of PPP	The purpose of the use case is to use data analytics and machine learning to assess viability and feasibility of PPP model for projects based on risk assessment, and collaboration potential between public and private sectors for infrastructure development			>12 Months
DED	AI for Economic Monitoring & Regulation	The purpose of the use case is to leverage business licensing data and historical trends to generate industry-insights and recommend the types of businesses to encourage or discourage based on market potential and job creation			6-12 Months
★ DED	AI for Business License	The purpose of the use case is to verify documents, identify potential issues, streamline inspections/fines process, collect financial statements and suggest necessary clarifications using AI.			6-12 Months
DGE - GovDigital	AI in project documentation and communication	The purpose of the use case is to utilize GenAI to automate the summaries of meetings, generate comprehensive status reports, streamline progress updates and provide templates for accurate and consistent project documentation, ensuring alignment among teams and stakeholders, the reason behind that is to try to address inefficiencies in project management and improve outcomes			6-12 Months

B ADGE PRIORITIZED USE CASES 6-12 MONTHS & >12 MONTHS (3/5)

ADGE Name	Use Case Title	Use Case Description	AI Impact	Feasibility	Time to Deploy
DGE - GovDigital	Digital Assistant	The purpose of the use case is to deploy a GenAI-enabled digital assistant to swiftly review knowledge management systems, analyze past tickets, and identify relevant components to assist in handling incoming requests, the reason behind that is to serve as a co-pilot for operators			6-12 Months
DGE - GovDigital	AI License Manager	The purpose of the use case is to leverage artificial intelligence algorithms to analyze software usage patterns, identify underutilized or unused licenses, and optimize license allocation across ADGEs, the reason behind that is to enable organizations to reduce costs and support overheads associated with unused or underutilized licenses			6-12 Months
DGE - GovAcademy	NLP based advisor	The purpose of the use case is to have a conversational platform that enables employees to engage in interactive learning experiences through natural language interaction			6-12 Months
★ DGE - GovAcademy	Content creation	The purpose of the use case is to leverage GenAI and create new training content or enhance existing training content (e.g., development of educational materials, resources, and activities)			6-12 Months
DGE - GovAcademy	AR/ VR driven programs	The purpose of the use case is to leverage augmented reality and virtual reality technologies to provide immersive learning experiences that blend digital content with the real world			>12 Months
DGE - GovProcurement	AI enabled Contract Management	The Purpose of the use case is to streamline the manual and error prone process of contract authoring, review, analysis, and management, and automates various aspects of the contract lifecycle reducing risks and improving contract compliance			6-12 Months
DGE - GovTalent	Professional Growth Pathway Creation	The purpose of the use case is to support employees and job seekers in developing a personalized plan, the reason behind that is to support their professional growth and skill development through a guided process based on set priorities, proven skills and other dimensions			6-12 Months
DGE - GovTalent	AI-Powered Skill Gap Analysis	The purpose of the use case is to use AI to analyze current employee skills, the reason behind that is to compare them to future job requirements, and recommend necessary training programs			6-12 Months
DMT	Smart City Platform	The purpose of this use case is to use data and sensors to optimize city operations, improve services like transportation and energy management, and enhance quality of life for residents through real-time data analysis and decision-making			>12 Months
DMT	Facial Recognition and Driver / Passenger Behaviour Analysis for PT and Taxi Safety and Fare Payment	The purpose of the use case is to analyze footage for driver fatigue, incidents, and automate passenger fare payment linked with registered accounts and detect safety issues for vulnerable passengers			6-12 Months
DMT	AI to Enhance Transportation Planning and Traffic Management	The purpose of this use case is to analyze vast datasets for predictive traffic flow, congestion prevention, and optimized transportation modelling and infrastructure planning			6-12 Months

B ADGE PRIORITIZED USE CASES 6-12 MONTHS & >12 MONTHS (4/5)

ADGE Name	Use Case Title	Use Case Description	AI Impact	Feasibility	Time to Deploy
DMT	Public Parking Management	The purpose of this use case is to optimize the management of public parking spaces. This includes real-time monitoring of parking availability, implementing dynamic pricing strategies, optimizing enforcement efforts, and providing convenient parking guidance to drivers			6-12 Months
DMT	Capture City Distortions through AI	The purpose of the use case is to utilize AI-driven methodologies to monitor and manage the ongoing expansion of urban infrastructure and green spaces effectively. This involves employing innovative techniques to identify, analyze, and address distortions or disruptions in the urban landscape, ensuring optimal utilization of resources and preservation of environmental sustainability			6-12 Months
DOE	Smart Grid Optimization	The purpose of the use case is to facilitate the management of electricity flow across the grid, enhancing efficiency by predicting demand patterns and integrating renewable energy sources smoothly			>12 Months
DOE	Carbon Capture Utilization and Storage	The purpose of this use case is to enhance the efficiency and effectiveness of carbon capture, utilization, and storage technologies. This will directly support DoE in reducing their carbon footprint effectively and economically, thereby aiding in meeting global climate targets and promoting environmental sustainability			>12 Months
DOE	Demand side Management: Forecasting Water and electricity consumption	The purpose of the use case is to forecast water and electricity consumption more accurately by utilizing predictive analytics and AI. This will in turn improve resource management and efficiency by predicting the demand fluctuations based on various factors like weather, time of day, and consumer behavior			>12 Months
DOE	Renewable energy forecasting	The purpose of the use case is to leverage AI to forecast demand for renewable energy sources such as solar and wind. Accurate forecasting will enable better integration of renewable energy into the power grid, optimize energy production, and enhance market participation			>12 Months
DOE	Infrastructure Predictive Maintenance	The purpose of this use case is to leverage AI to monitor infrastructure and equipment performance through predefined measures. The goal is to identify anomalies, predict maintenance needs, and schedule improvements proactively to minimize downtime and prevent failures.			6-12 Months
DOH	Physician Clinical LLM	The purpose of this use case is to provide physicians with a search tool within “Malaffi” to quickly access active patient information, for example anomalies in a patient’s latest blood test results			6-12 Months
DOH	AI Medical Imaging	The purpose of this use case is to significantly improve early detection and accuracy of diagnosing diseases through the analysis of X-ray and CT scans across a range of diseases (e.g., lung disease). This capability not only speeds up the diagnostic process but also enhances the precision of the results, potentially leading to early interventions and better patient outcomes			6-12 Months
SCAD	Livability - GenAI	The Purpose of the use case is to evaluate and enhance urban livability factors including safety, accessibility, and environmental quality, facilitating smarter urban planning and improved living conditions			6-12 Months
SCAD	Environment Pollution	The Purpose of the use case is to utilize the Economic Complexity Index model leveraging AI and data analytics to monitor, analyze, and mitigate sources of pollution, enabling data-driven decision-making for environmental protection and sustainability initiatives			6-12 Months

B ADGE PRIORITIZED USE CASES 6-12 MONTHS & >12 MONTHS (5/5)

ADGE Name	Use Case Title	Use Case Description	AI Impact	Feasibility	Time to Deploy
SCAD	Forecasting GDP up to 5 years	The Purpose of the use case is to forecast GDP for the next 5 years across various sectors in Abu Dhabi. This initiative aims to predict economic trends, assist in resource allocation, and support decision-making processes that influence the region's development and economic stability			6-12 Months
SCAD	Population and Migration Forecast up to 5 years	The Purpose of the use case is to forecast population dynamics and migration patterns over the next 5 years, aiming to understand demographic shifts and their implications on urban planning, workforce development, and social services			6-12 Months
SCAD	Domestic Agriculture	The Purpose of the use case is to predict agricultural production levels, helping to enhance food security and optimize resource management across domestic markets. This forecasting aims to support agricultural planning and sustainable farming practices			>12 Months

To enable AI adoption across whole-of-government, GovAI will offer key enablers

